

# Cross-Border Flows of Construction:

## The Myth of the Non-Tradable Sector\*

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### Abstract

Construction is conventionally treated as the textbook example of a non-tradable sector. This paper challenges that view. Using data on global construction firms, World Bank projects, and trade in construction inputs, we document three facts: (i) Construction is a global sector, but its cross-border flows are almost invisible in standard data; (2) China has become the dominant global construction power, simultaneously displacing rich-country incumbents in services and goods; and (3) Low-income countries face a binding construction capacity constraint, they can execute small projects domestically, but large infrastructure requires foreign expertise. We develop a model of the decision to produce infrastructure projects. Each project's engineering can be supplied by domestic engineers or imported from global contractors. The model rationalizes the binding capacity constraint of low-income countries, predicts which countries import engineering services, and, with learning-by-doing, delivers a dynamic cost of using foreign engineers.

**JEL:** F14, F23, L74, O18, O24.    **Keywords:** construction, trade in services, non-tradable sectors, infrastructure, China, development.

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## 1. Introduction

While construction is a large, important industry in every country, it is rarely included in discussions of globalization, international trade, or foreign investment. Instead, it is typically viewed as the canonical example of a non-tradable sector with the final product, such as a road, power plant, or airport, produced where it is used. Standard balance-of-payments statistics appear to confirm this intuition: recorded cross-border construction services amounted to roughly \$89 billion in 2024 representing well under 1 percent of total services trade.<sup>1</sup>

These statistics miss almost all the cross-border activity in construction. The 250 largest international contracting firms, ranked by Engineering News-Record (ENR) on the basis of revenues earned *outside* their home countries, collectively reported over \$500 billion in international revenues in 2024 amounting to one-and-a-half times the value of global trade in aviation equipment and comparable to that in automotive parts. The value of imported goods used on construction sites, i.e. machinery, equipment, and materials, was at least \$660 billion, and as much as \$1.8 trillion when including the full range of large-project inputs. The gap between official statistics and economic reality is large because the dominant delivery model for international construction is GATS Mode 3: the firm establishes a local affiliate or deploys a team of engineers to the host country, whose revenues appear neither as services exports in home-country balance-of-payments accounts nor reliably as foreign direct investment, given the project-based and often temporary nature of the activity.

What is actually being traded? While structures themselves cannot cross borders, a key ingredient for building them can: engineering services. The capacity

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<sup>1</sup>Outbound direct investment in construction is comparably low, representing less than 1% of aggregate outbound FDI for the US (\$8B) and EU (€78B) and under 2% for China (\$56B). See Appendix Table A.3

to plan, design, organize, and execute a complex infrastructure project is non-rival within a firm as the same systems and accumulated expertise can be deployed on multiple projects in multiple countries, but it is embodied in people, specifically in the highly skilled engineers and project managers of global construction firms. It has long been recognized that this capacity is scarce in low-income countries. International competitive procurement was introduced by the World Bank in 1951, as part of a broader effort in which the “*Bank recognized that its borrowers often lacked the skills to conceive, design, and carry out the development projects*” (King, 1975). The resulting gap between the engineering capacity of rich and poor countries has not closed and drives an important part of the global market for construction services.

This paper studies the extent, evolution, and implications of cross-border flows in construction. We assemble three distinct datasets. First, we digitize ENR annual top-list publications from 1989 to 2025, yielding approximately 7,200 firm-year observations and 72,000 firm-year-destination observations recording each firm’s home country, revenues, sectors of activity, and countries of operation. Second, we use World Bank Investment Project Financing (IPF) procurement records for 2020-2026, covering roughly 55,000 works (construction) contracts with known borrower countries, supplier identities and nationalities, and contract values. Third, we classify all 5,219 six-digit HS codes in the BACI trade database (2002 classification) into whether they are core construction inputs, multi-sector inputs with significant construction use, or unrelated to construction, and further classify each construction-relevant code by whether its primary application is in large-project infrastructure or in residential and small-scale construction.

Using these data, we document three facts. First, cross-border flows in construction are *large* but almost entirely invisible in standard trade statistics. The top international contractors work in essentially every country in the world. Trade in large-project construction inputs alone reached approximately \$1.8 trillion in 2024. The official \$89 billion figure understates the true scale of cross-border construction

activity by more than an order of magnitude.

Second, there has been a *dramatic shift in who provides global infrastructure*, with China emerging as the dominant supplier across both services and goods. China's share of aggregate top international contractor revenues rose from under 1 percent in 1989 to 25 percent by 2024. In sub-Saharan Africa, Chinese contractors went from serving roughly 5 percent of the market to approximately 60 percent. China's share of global large-project construction input exports tripled between 2002 and 2024, rising from 6 to 24 percent, outpacing its already-exceptional overall export growth.

Third, *low-income countries face a binding construction capacity constraint*. Not a single low-income country is home to a top-250 international contractor; the poorest World Bank borrowers import more than 60 percent of their works contract value from foreign firms.

The second contribution of the paper is to analyze the implications of these cross-border flows for low-income countries. Imports of construction services have the potential to remove domestic bottlenecks, leading to higher infrastructure stocks and subsequent growth. However, such imported services may stifle the development of local expertise, since learning-by-doing in engineering is an important channel of capability accumulation. The government therefore faces a key trade-off. To analyze it, we develop a model of infrastructure provision with four key features: infrastructure is a form of capital; engineering services are tradable inputs that are complementary to non-tradable local labor; demand for large, complex projects stems from the government; and importing engineering expertise from abroad has a dynamic cost because of a learning-by-doing externality. The model yields tight predictions about which countries import engineering services and allows us to quantify the welfare consequences of China's rise.

**Related literature.** By studying cross-border flows in construction we relate to the literature on the production of structures in low-income countries ([Collier et al.](#),

2016; Kirchberger, 2020) and on public procurement of infrastructure (Bosio et al., 2022). Our findings connect to the literature on U.S. construction-sector productivity stagnation (Goolsbee and Syverson, 2023; D’Amico et al., 2024): the evidence here suggests the United States has also lost its position as a global leader in large-scale infrastructure. We relate to the literature on global firms and multinational production (Bernard et al., 2018; Conconi et al., 2016; Antràs et al., 2023), extending it to a sector characterized by Mode 3 delivery and project-based activity. Recent work on joint ventures and technology transfer in construction (Choi et al., 2025) and on trade in non-tradables (Muñoz, 2024) is closely related. On the services side, we contribute to the literature on trade in services (Francois and Hoekman, 2010; Jensen, 2011; Hsieh and Rossi-Hansberg, 2023), focusing on a Mode 3 sector that is systematically under-recorded in standard statistics. Finally, our paper relates to the trade-in-capital-goods literature (Eaton and Kortum, 2001; Hsieh and Klenow, 2007), connecting patterns of cross-country specialization in construction to the broader literature on how trade in investment goods affects development.

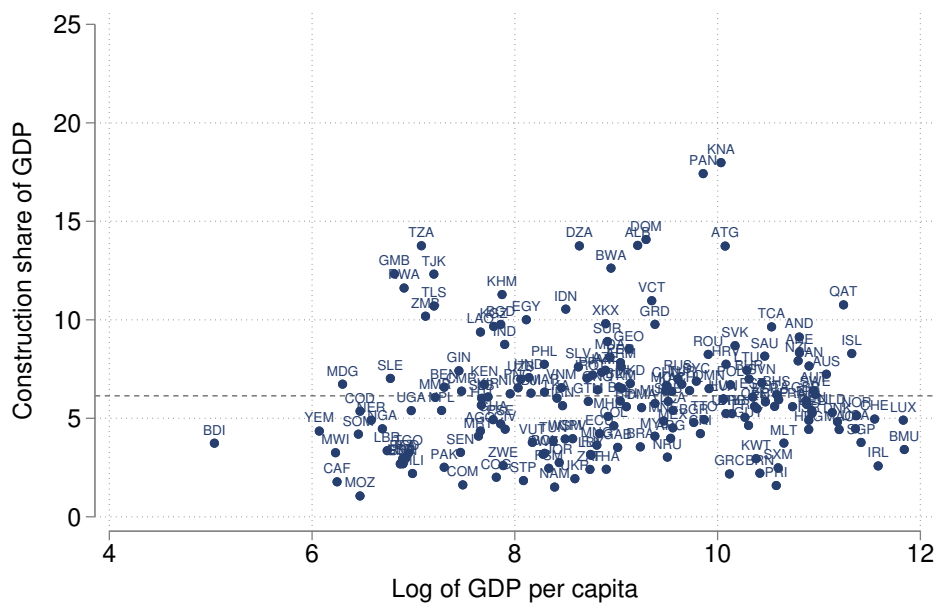
The paper proceeds as follows. Section 2 provides background on the size and structure of the global construction sector. Section 3 describes the three data sources. Section 4 documents the three facts. Section 5 develops the theoretical framework including responses to china’s rise and the learning-by-doing extension. In Section P we analyze the welfare consequences of taxing foreign engineers to enhance learning-by-doing. Section 7 concludes.

## **2. Construction in the Global Economy**

### **2.1. Size and stability**

Construction is a large sector in every country. From 2001–2024, the typical country had a construction GDP share between 4 and 7 percent, with a mean close to 6 percent, corresponding to approximately \$7 trillion in value added and \$14 trillion in gross revenues globally in 2024, see Figure A.2. The distribution has been

Figure 1: Construction Share and Level of Development



Note: Cross-section scatter of the construction share in GDP against log per capita GDP. Each point is a country.

remarkably stable over time suggesting that demand for structures and infrastructure scales proportionally with income.

Figure 1 examines whether the sector's importance varies systematically with income. The scatter plot shows no systematic relationship between the construction share and log per capita GDP. Construction is as important, in share terms, in low-income countries as in high-income ones. Because total GDP rises steeply with development, the *level* of construction activity is far larger in rich countries, but the sector plays an equally central structural role everywhere.

## 2.2. What is traded: engineering services

While the final output of construction cannot be traded, a crucial ingredient can: engineering services. Large-scale infrastructure projects require the capacity to plan and design a complex project, to organize procurement and logistics, to schedule thousands of activities, and to supervise construction to demanding stan-

dards. This know-how is, in a meaningful sense, non-rival within the firm: the same systems, procedures, and accumulated experience can be deployed simultaneously on multiple projects in multiple countries. It is, however, embodied in the specialized engineers and project managers of global construction firms, and in particular in the highly skilled personnel who are sent to host countries to execute individual projects.<sup>2</sup>

The resulting market is one in which rich-country firms with accumulated engineering expertise supply services to governments that lack domestic capacity, precisely the pattern the World Bank identified when it introduced international competitive procurement in 1951. By 1964, firms from just four rich countries, the United States, the United Kingdom, Germany, and France, accounted for 55.6 percent of total World Bank/IDA contract value for goods and services. As we document in Section 4, this structure persists today, though its composition has changed dramatically.

### **2.3. Why standard data misses these flows**

The reason cross-border construction flows are so poorly recorded is that they are delivered primarily through GATS Mode 3, in which the supplying firm establishes a local affiliate or project office in the host country. Revenue earned by that affiliate does not appear as a services export in the home country's balance-of-payments construction accounts; it is recorded, if at all, as income of a foreign-owned enterprise, which may or may not surface in FDI income flows. For short-duration construction projects even the FDI recording may be incomplete. The balance-of-payments figure of \$89 billion captures primarily Mode 1 and Mode 4 flows (direct cross-border supply and movement of natural persons), which are a small fraction of the total.

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<sup>2</sup>We abstract from housing construction, which is typically informal in low-income countries and operates at a different scale with a different production function. Non-residential buildings and civil engineering account for about 60 percent of total construction expenditures globally (ICP, 2017).

### 3. Data

One key challenge in studying cross-border flows in construction is that there is no single readily available dataset; such flows are not captured by standard trade or FDI data. We use three complementary data sources. Appendix H provides additional details on construction and measurement.

#### 3.1. Top international contractors: Engineering News-Record

Engineering News-Record (ENR) began publication in 1840 and has collected intelligence on the global construction industry continuously since then. It is regarded as a reliable source of industry data (Lu, 2014). Since 1989 it has published an annual ranking of top international contractors, defined by revenue earned *outside* the home country. The list covered 225 firms through 2012 and has covered 250 firms from 2013 onward.<sup>3</sup>

We digitize all available editions from 1989 to 2025. For each firm-year we record the firm's name and rank, the city and country of headquarters, total and international revenues, and the main sectors of activity (general building, manufacturing, transportation, power, water supply and sewerage, petroleum, telecoms, and others). A separate supplement lists, for each contractor, the countries in which it was active in that year; these supplements are available from 1989–1998 and continuously from 2005 to 2025.

The resulting dataset contains approximately 7,200 firm-year observations and 72,000 firm-year-destination observations. The average international revenue share, the fraction of a firm's total revenue earned outside its home country, is about 50 percent, confirming that cross-border activity is central to the business model of these firms.

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<sup>3</sup>Throughout the paper we limit the data to firms ranked 225 or higher.

### 3.2. World Bank investment project financing

Our second data source is the World Bank IPF procurement database covering contracts awarded in 2020–2026 ([World Bank Group, 2020](#)). For each contract we observe the project name and ID, the borrower country, the name and registered country of the supplier, and the contract amount.<sup>4</sup> There are four procurement categories: works, goods, consultant services, and non-consultant services. We focus on *works*, corresponding to construction contracts proper, leaving approximately 55,500 contracts. The sample is restricted to World Bank clients, which excludes high-income countries.

### 3.3. Trade in construction-related goods: BACI

Our third data source is the BACI bilateral trade dataset compiled by CEPII from Comtrade data ([Gaulier and Zignago, 2010](#)), covering 2002–2024 at the HS6 level under the HS2002 classification. Because no standard classification identifies construction inputs, we construct an original categorization using LLMs to assign each of the 5,219 six-digit HS codes to one of four mutually exclusive groups:

**Group A (Core construction):** Construction is the dominant or exclusive end-use (3.9% of HS codes). Examples: excavators and front-loaders (HS 842951), high-voltage cable above 1,000V (HS 854460), railway track rails (HS 730210).

**Group B (Multi-sector):** Construction is a significant but not dominant end-use (28.0% of codes). Examples: steam boilers (HS 840290), aluminium window frames (HS 761010), polyethylene sheets (HS 392010).

**Group C (Noisy):** Construction use is possible but uncertain.

**Group D (None):** No construction end-use, e.g. aircraft, pharmaceuticals, apparel, consumer electronics, and the large majority of traded goods.

For all Group A and Group B goods we apply a further binary classification:

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<sup>4</sup>The place of registration is not necessarily the headquarters location. Joint-venture contract values are divided equally across consortium members; subcontracting flows are not recorded, so import shares we compute likely *understate* the true foreign share.

Table 1: Classification Examples: Large-Project Construction Inputs

| HS code                                                                   | Product                                | Rationale                                                          |
|---------------------------------------------------------------------------|----------------------------------------|--------------------------------------------------------------------|
| <i>Group A, large project = 1 (core construction, infrastructure use)</i> |                                        |                                                                    |
| 842951                                                                    | Excavators and front-end loaders       | Used on large civil and mining sites; rarely in residential builds |
| 854460                                                                    | High-voltage electric cable (>1,000 V) | Power-transmission infrastructure; not used in residential wiring  |
| 730210                                                                    | Railway track rails                    | Exclusively large transport infrastructure                         |
| <i>Group A, large project = 0 (core construction, residential use)</i>    |                                        |                                                                    |
| 680911                                                                    | Plasterboard (gypsum wallboard)        | Heavily used in residential construction                           |
| 690721                                                                    | Ceramic floor and wall tiles           | Construction finishing; demand overwhelmingly residential          |
| <i>Group B, large project = 1 (multi-sector, infrastructure use)</i>      |                                        |                                                                    |
| 840290                                                                    | Steam boilers and parts                | Primarily power-plant construction; also industrial replacement    |
| 732010                                                                    | Leaf springs for heavy structures      | Bridge and industrial-frame components                             |
| <i>Group B, large project = 0 (multi-sector, residential use)</i>         |                                        |                                                                    |
| 392010                                                                    | Polyethylene sheets and film           | Damp-proof membrane in construction; also packaging                |
| 761010                                                                    | Aluminium doors, windows and frames    | Residential and small-commercial volumes dominate                  |

*Note:* LP = large-project indicator. Classification performed using large language models applied to all 5,219 six-digit HS2002 codes.

whether the primary construction end-use is in **large-project infrastructure** (civil engineering, power generation, transportation) or in other construction (primarily residential). Approximately 7.9 percent of all HS codes carry this **Large Project (LP)** designation. Table 1 provides examples. In 2024, trade in LP goods amounted to approximately \$1.8 trillion, or roughly 8 percent of global merchandise trade.

## 4. Three Facts on Cross-Border Construction Flows

This section documents three facts that together overturn the conventional view of construction as non-tradable. The first fact establishes that the scale of cross-border flows is large and systematic, but almost entirely invisible in standard statistics. The second documents a dramatic supply-side transformation driven by China. The third shows that low-income countries are fundamentally dependent on foreign expertise for large infrastructure projects.

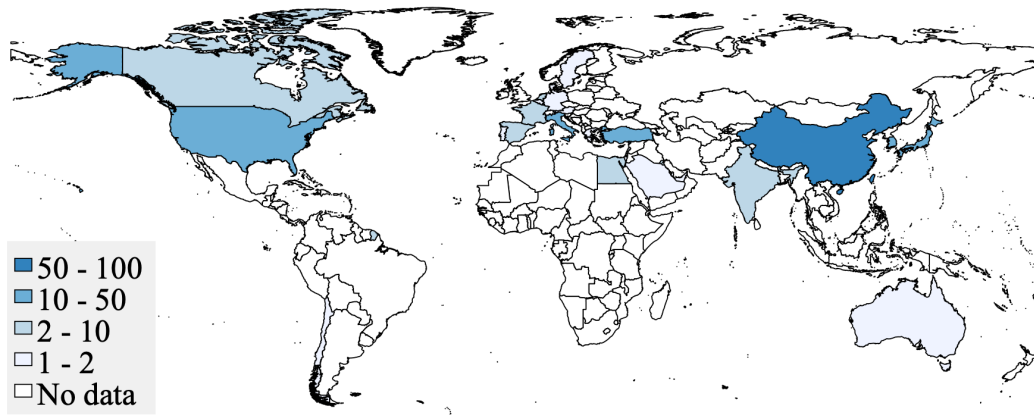
### 4.1. Fact 1: Cross-border construction flows are large and systematically under-measured

The official balance-of-payments figure for construction services trade, \$89 billion in 2024, misses the dominant channel through which international construction is transacted. The ENR data make this plain. Figure 2 maps the number of top-250 international contractors headquartered in each country in the most recent year. The supply side is highly concentrated: nearly all global contractors originate from a small set of high-income countries, such as European economies, the United States, Japan, and South Korea, together with China. Eighty percent of contractors come from just seven countries.

Figure 3 confirms the income gradient formally. The number of global construction firms per ten million inhabitants rises steeply with per capita income. The large majority of countries, especially at lower income levels, have no firms on the top-250 list.

The firms on the ENR list work in essentially every country in the world. Figure 4 maps the number of top-250 firms active in each country. Very few countries appear with no foreign contractor presence; most of these reflect data gaps rather than genuine absence. High-income and low-income countries alike contract with global construction firms, and the United States itself hosted more than 50 foreign contractors in recent years.

Figure 2: Origins of the Top 250 International Contractors, 2024



*Note:* Number of top-250 international contractors headquartered in each country in 2024.  
*Source:* ENR.

The goods-trade dimension of cross-border construction flows is similarly large. Appendix Figure A.5 shows our construction-input categories as a share of global merchandise trade from 2002 to 2024. The large-project (LP) category alone has been stable at roughly 8 percent of global merchandise trade throughout the period, reaching approximately \$1.8 trillion in 2024, a figure that dwarfs the official BOP estimate of construction services trade.

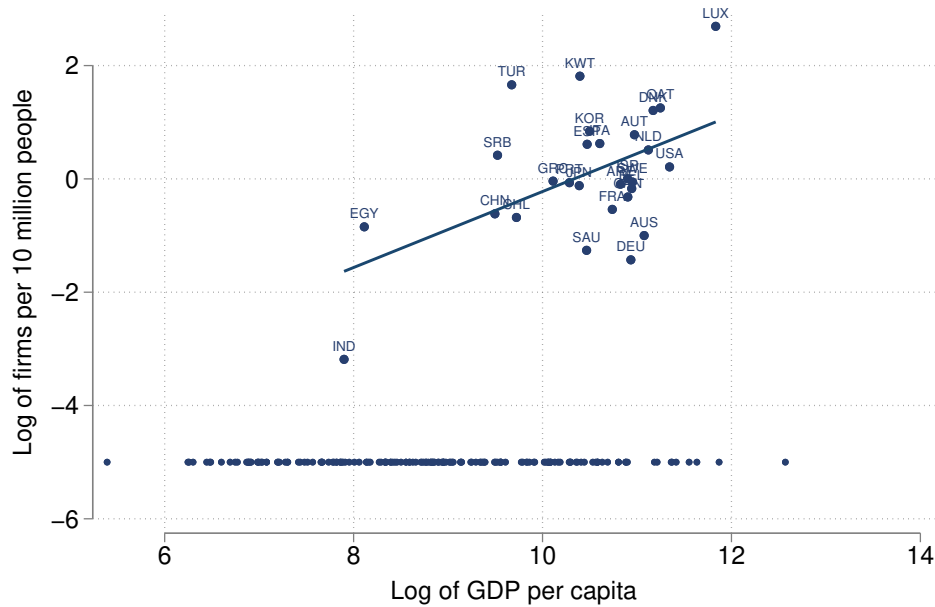
Taken together, the ENR contractor revenues (\$500B+), the imported goods flows (\$1.8T in LP inputs), and the global presence of the top 250 firms establish substantial cross-border nature of the global construction sector.

#### **4.2. Fact 2: China has become the dominant global construction power, displacing rich-country incumbents in both services and goods**

Figure 5 plots aggregate international revenues by home region of the contractor from 2000-2024. The figure reveals two transformations that together constitute one of the most significant shifts in global service trade of the past three decades.

The first is the rise of China. Chinese firms were essentially absent from the

Figure 3: Income per Capita and Global Construction Firms

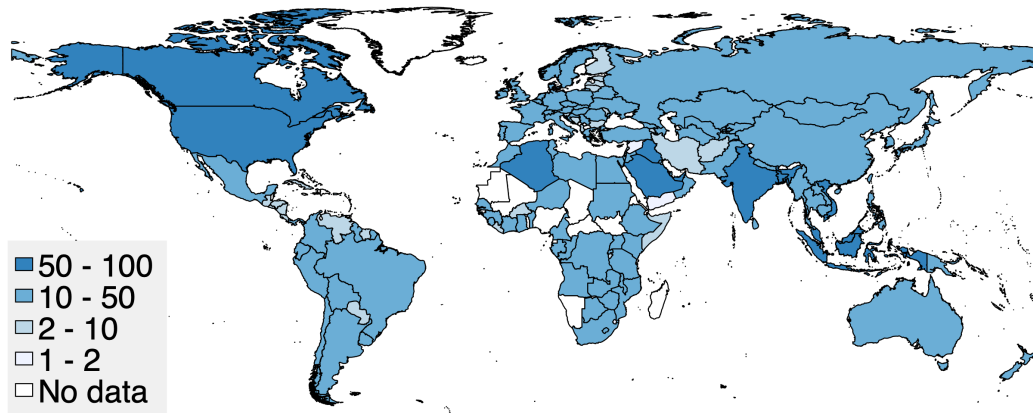


*Note:* Number of top-250 international contractors per 10 million population against log per capita GDP. -5 indicates no top-250 contractors. Source: ENR; World Development Indicators.

ENR rankings in the late 1980s, with a share of international revenues below 1 percent. By 2024 that share had reached approximately 25 percent, making China the second-largest supplier of international construction services behind Europe as a whole. Until 2011, all of the top 10 international contractors came from high-income countries; by 2024 seven of the top 20 firms are Chinese.

The second is the sharp decline of the United States. U.S. firms peaked at close to 28 percent of global international construction revenues in the early 1990s. By 2024 the share had fallen to approximately 6 percent. The number of U.S. firms on the ENR top-250 list fell from above 80 at its peak to around 30 today. This finding connects to the broader literature on U.S. construction productivity stagnation (Goolsbee and Syverson, 2023): the evidence here suggests that the United States has not only experienced weak domestic construction productivity growth but has also lost its position as a global leader in the provision of large-scale infrastructure.

Figure 4: Destinations of the Top 250 International Contractors, 2024

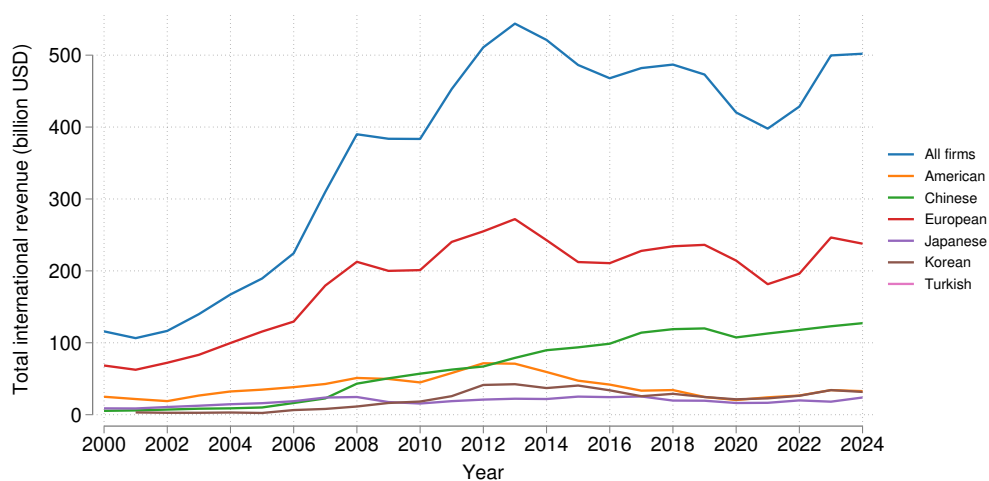


*Note:* Number of top-250 international contractors active in each country in 2024. Source: ENR.

China's rise is most striking in sub-Saharan Africa. Figure 6 shows market shares by home region in the African market over time. Chinese contractors held roughly 5 percent of the market around 2000; by 2024 that share had grown to approximately 60 percent, largely displacing European and American firms that previously dominated. Europe's share fell from about 50 percent to roughly 20 percent over the same period; the U.S. share contracted from around 20 percent to below 5 percent.

China's dominance extends beyond contractor services to construction-related goods. Figure 7 shows China's share of large-project construction input imports for sub-Saharan Africa and South Asia, the two fastest-growing regions for such trade. In sub-Saharan Africa, China's share rose from roughly 7 percent in 2002 to close to 40 percent in 2024; in South Asia, the increase was from around 10 percent to approximately 40 percent. Both regions have seen the fastest growth in LP input imports globally, sub-Saharan Africa expanded roughly sevenfold relative to 2002 and South Asia roughly twelvefold, and China has captured an ever-larger share of that growth.

Figure 5: Aggregate International Revenues by Home Region



*Note:* Nominal annual international revenues for top-250 contractors, by home region.  
Source: ENR, 2000–2024.

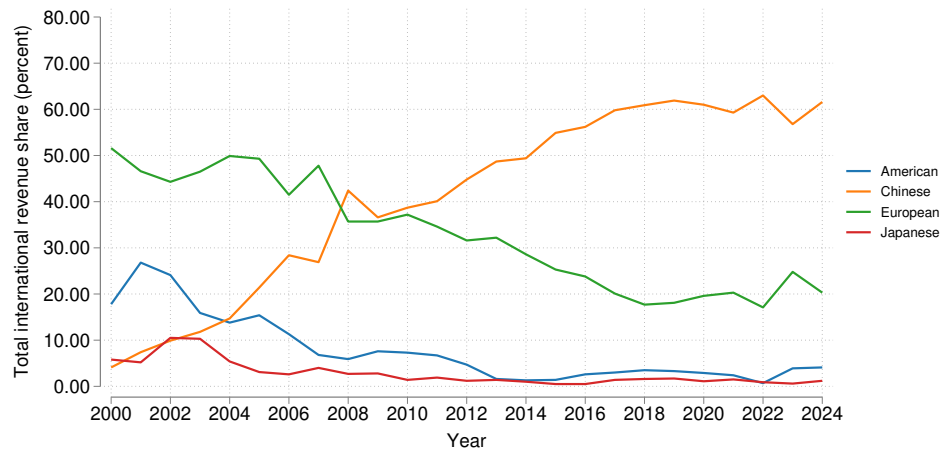
At the global level, Appendix Figure A.10 reveals that China’s rise in construction inputs is even more remarkable than its overall export performance. While China’s share of world merchandise exports roughly doubled from 2002 to 2024, from about 7.5 to 15 percent, its share of global large-project construction input exports more than tripled, rising from about 6 to 24 percent. China has developed a comparative advantage in construction inputs that is even more pronounced than its formidable general manufacturing export specialization.

#### 4.3. Fact 3: Low-income countries face a binding construction capacity constraint, large infrastructure requires foreign expertise

As Figure 2 already makes clear, not a single low-income country is home to a top-250 international contractor. Lower-middle-income countries play only a negligible role. The supply of global construction expertise is, in this sense, entirely absent from the poorest parts of the world. Our third fact characterizes what this means for the demand side.

Figure 8 plots the share of total works contract value provided by foreign sup-

Figure 6: Market Shares in Sub-Saharan Africa

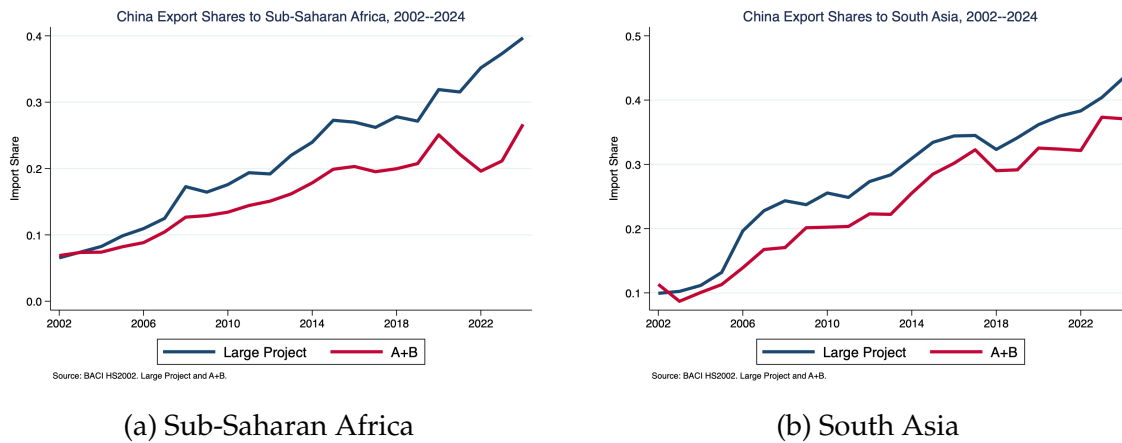


Note: Share of total international revenues in sub-Saharan Africa by home region of contractor, 2001–2024. Source: ENR.

pliers against per capita income for each World Bank borrower country. The relationship is strongly negative. The poorest countries import the largest fraction of their construction contracts: many low-income countries import more than 60 percent of the contract value of World Bank-financed works, while middle-income borrowers typically import around 30 percent. There is substantial heterogeneity even within income groups, Nigeria and Ghana import 40 percent or less, compared with Niger and Côte d'Ivoire, who import more than 60 percent, suggesting that institutional capacity and existing contractor relationships matter beyond income alone.

Figure 9 turns to the *scale* of imported versus domestically executed projects, providing the sharpest evidence for the capacity constraint interpretation. Panel (a) plots the distribution of contract values for foreign and domestic suppliers separately. The two distributions are essentially non-overlapping. The median imported contract exceeds \$1 million; the median domestic contract lies below \$50,000. The gap is more than an order of magnitude. Domestic construction firms in low-income World Bank clients can execute small-scale works, but are entirely absent

Figure 7: China's Share of Large-Project Construction Input Imports



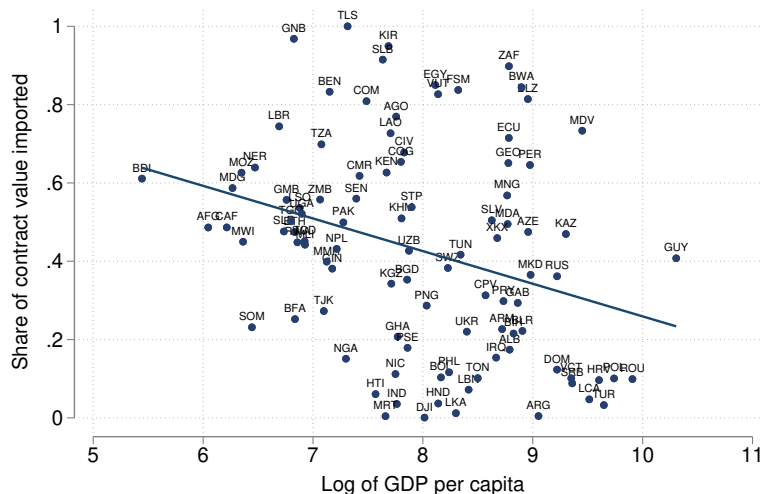
*Note:* China's share of large-project (LP) construction input imports, 2002–2024. Source: BACI CEPII.

from large infrastructure contracts.

Panel (b) reinforces the interpretation. The average contract value for foreign-supplier projects is broadly similar across poor and lower-middle-income countries, in the range of \$5 to \$10 million, suggesting that the scale of projects that require foreign expertise does not vary much with income. In contrast, average domestic contract size is substantially smaller in the poorest countries and rises with income. Domestic construction capabilities accumulate with development; in the poorest countries, the ability to execute large projects is essentially absent.

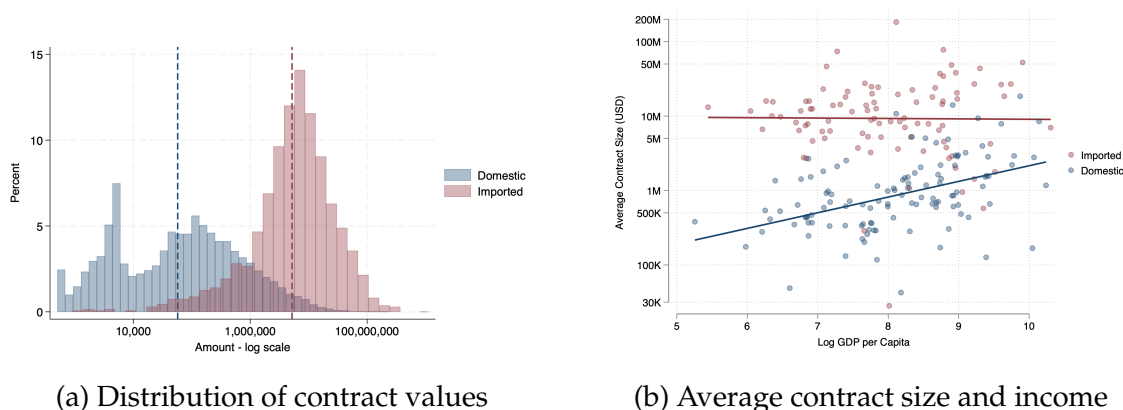
**Summary.** The three facts together establish a coherent picture: construction is a globally traded sector whose flows are systematically invisible in standard data; the supply side has been transformed by the rise of China at the expense of rich-country incumbents, particularly in the developing world; and low-income countries depend critically on foreign contractors for any large-scale infrastructure, revealing a binding capacity constraint with direct implications for development.

Figure 8: Import Share of Construction and Development



*Note:* Share of total works contract value awarded to foreign suppliers against log per capita GDP, World Bank borrower countries, 2020–2026. Per capita GDP from WDI; 2024 values used for 2025–2026. Source: World Bank IPF; WDI.

Figure 9: Scale of Imported and Domestic Construction Projects



*Note:* Panel (a): distribution of contract values (log scale) for foreign and domestic suppliers. Panel (b): average contract size against per capita GDP, for foreign and domestic suppliers, among low-income and lower-middle-income borrowers. Source: World Bank IPF.

## 5. Theory

In this section we provide a simple framework to analyze the implications of international flows in infrastructure projects. Four features of the data discipline the model. First, infrastructure is built from projects that vary enormously in size, and the size distribution of imported and domestically executed projects barely overlaps. Second, the supply of engineering expertise is highly concentrated in rich countries and the poorest borrowers import more than 60 percent of their works contract value. Third, even capable, high-income countries continue to import specialized infrastructure: the United States alone hosts more than fifty foreign contractors, concentrated in technologically demanding work. Fourth, China's entry has sharply reduced the price of imported engineering and reallocated the global market toward imports.

The model is built to speak to these facts. Its central object is the *sourcing* decision for a project's engineering, domestic or foreign, and its central mechanism is a per-project fixed cost of engaging a foreign engineering team. That fixed cost makes imports relatively cheap for large projects and relatively expensive for small ones, producing a size threshold above which projects are imported. A country's position in this threshold, and hence its import share, is governed by its domestic engineering capability and by the world price of engineering. Allowing projects to differ in technological complexity, on which global firms hold a frontier advantage, adds a second margin: the most complex projects are imported regardless of size, so even highly capable countries retain a frontier tail of imports.

In this section we lay out the environment and then characterizing the static sorting equilibrium. After introducing country heterogeneity, we add project complexity and the frontier tail of rich-country imports.

### 5.1. Environment

We start out with a static model to highlight the main tradeoffs. In Section [O](#) below we extend our analysis to a dynamic setting to highlight the role of learning-by-doing externalities and the long-lived nature of infrastructural capital.

**Final good and preferences.** A representative economy is endowed with  $L$  units of labor and produces a single final good from production labor  $\ell_P$  and a stock of infrastructure  $I$ ,

$$Y = \left( \lambda (A \ell_P)^{\frac{\sigma-1}{\sigma}} + (1 - \lambda) I^{\frac{\sigma-1}{\sigma}} \right)^{\frac{\sigma}{\sigma-1}}, \quad (1)$$

where  $A$  is labor productivity and  $\sigma$  the elasticity of substitution between labor and infrastructure. The amount of infrastructure is given by

$$I = I_0 + \mathcal{B}, \quad (2)$$

where  $I_0$  is the initial stock of infrastructure in place and  $\mathcal{B}$  denotes new construction. In the static model of this section, we take  $I_0$  in (2) as given. In the dynamic model in Section [O](#), (2) will take the form of a dynamic accumulation equation. Throughout we focus on the case  $\sigma < 1$ , so that labor and infrastructure are complements and a shortage of infrastructure is a binding bottleneck. Note that this implies that an increase in  $A_t$  *increases* the demand for infrastructure because of the complementarity.

In terms of preferences, we assume that there is a representative household who consumes the final good and has log utility;  $u(c) = \ln c$ . Labor is supplied inelastically.

**Infrastructure as a portfolio of projects.** Infrastructure is not a homogeneous stock but the sum of completed discrete projects, i.e., roads, ports, and power plants, that differ in scale. We assume that at any point in time there is a unit

mass of potential projects indexed by size  $s > 0$ , with twice-differentiable c.d.f.  $H(s)$  and density  $h(s)$ . Completing a project of size  $s$  adds  $s$  units to the stock of infrastructure. Letting  $\mathcal{N} \subseteq (0, \infty)$  denote the endogenous set of completed projects, total infrastructure investment is given by

$$\mathcal{B} = \int_{\mathcal{N}} s dH(s). \quad (3)$$

Infrastructure requires two factors of production: construction workers and engineering services. Completing a project of size  $s$  requires  $\theta^{-1}s$  units of *construction labor*  $\ell_C$ . Hence,  $\theta$  governs the productivity of construction workers. Construction labor is always provided domestically and is used by every completed project regardless of how it is engineered: it is engineering, not construction, that is tradable. Total demand for construction workers is therefore given by

$$\ell_C = \int_{\mathcal{N}} \frac{1}{\theta} s dH(s) = \frac{1}{\theta} \mathcal{B}. \quad (4)$$

In addition to construction workers, every project requires engineering, including design, procurement, scheduling, and supervision. We assume that engineering services are also proportional to the project's size: completing project  $s$  requires  $s$  efficiency units of engineering. Crucially, engineering services are tradable and can either be supplied by domestic engineers or sourced from abroad. However, for a given project, engineering is supplied by a *single* source, reflecting that a project is designed and managed by one lead engineering team.

- *Domestic engineers.* Local engineers have efficiency  $q > 0$ . Delivering  $s$  efficiency units therefore requires  $q^{-1}s$  local engineers and therefore costs  $\frac{w}{q}s$ , where  $w$  is the price of labor. Lower engineering capabilities raise the effective unit cost  $w/q$ .
- *Foreign engineers.* Engineering services can also be sourced from abroad. A

global contractor supplies engineering at the world price  $p$  per efficiency unit. To mobilize a foreign team for the project, there is a fixed cost  $\varphi > 0$  to travel, setup, contract, and coordinate on-site. Importing the engineering for project  $s$  thus costs  $p(s + \varphi)$ .

With a per-project fixed cost, large and small projects sort differently: foreign engineers have a comparative advantage in large projects. Both the amount of investment  $\mathcal{B}$  and the composition of project size therefore depends on  $q$ ,  $w$ ,  $p$ , and  $\varphi$ .

More specifically, given that domestic engineering is provided with constant returns to scale, the total demand for domestic engineers is given by

$$\ell_E = \int_{\mathcal{N}} \frac{1}{q} s dH(s) = \frac{1}{q} \mathcal{B}^D, \quad (5)$$

where  $\mathcal{B}^D$  denotes the amount of construction that is done domestically. Note that the size distribution of domestic projects is indeterminate.

For foreign engineers, the calculus is different. Because fixed costs do not scale in size, the projects done by imported engineering services are *positively selected on size*: letting  $\mathcal{N}^F$  the set of projects with foreign engineers, we have

$$\mathcal{N}^F = \{s | s > s^F\}, \quad (6)$$

where  $s^F$  is a size cutoff of projects with foreign engineers. Intuitively, to minimize the fixed costs of foreign engineers, one would always start with the largest projects. To build a measure  $\mathcal{B}^F$  of foreign infrastructure wherefore requires  $x(\mathcal{B}^F)$  units of foreign engineering services:

$$x(\mathcal{B}^F) = \int_{s^F(\mathcal{B}^F)}^{\infty} s dH(s) = \mathcal{B}^F, \quad (7)$$

where  $s^F(\mathcal{B}^F)$  is the size cutoff for foreign-run projects to build  $\mathcal{B}^F$  units of infras-

trucure.

**The Costs of Infrastructure** We can now derive the cost function of infrastructure. Let  $C^D(\mathcal{B})$  denote the total cost to produce  $\mathcal{B}$  units of infrastructure with domestic engineers and domestic construction workers. These costs are given by

$$C^D(\mathcal{B}) = w (\ell_C(\mathcal{B}) + \ell_E(\mathcal{B})) = \left( \frac{1}{q} + \frac{1}{\theta} \right) w \mathcal{B}, \quad (8)$$

reflecting the fact that one needs  $\frac{\mathcal{B}}{\theta}$  construction workers and  $\frac{\mathcal{B}}{q}$  engineers, both of which have a price of  $w$ .

By contrast, the total costs to produce  $\mathcal{B}$  with foreign engineers (and domestic construction workers) are given by

$$C^F(\mathcal{B}) = w \frac{1}{\theta} \mathcal{B} + p (\mathcal{B} + (1 - H(s^F(\mathcal{B}))) \varphi), \quad (9)$$

where  $s^F(\mathcal{B})$  is defined in (7) and  $(1 - H(s^F(\mathcal{B})))$  is the mass of projects that are done with foreign engineers. A higher level of spending  $\mathcal{B}$  *increases* the mass of projects by reducing the cutoff  $s^F(\mathcal{B})$  and drives up the fixed costs.

A particularly nice expression for (9) arises if  $s$  is Pareto distributed on  $[s_0, \infty]$  with shape  $\gamma \geq 1$ , i.e.

$$H(k) = \mathbf{P}(s \leq k) = 1 - \left( \frac{s_0}{k} \right)^\gamma. \quad (10)$$

For that case, we have  $1 - H(s^F(\mathcal{B})) = \left( \frac{s_0}{s^F(\mathcal{B})} \right)^\gamma$  and  $s^F(\mathcal{B})$  solves the equation

$$\mathcal{B} = \int_{s^F(\mathcal{B})}^{\infty} s dH(s) = \left( \frac{s_0}{s^F(\mathcal{B})} \right)^\gamma \frac{\gamma}{\gamma - 1} s^F(\mathcal{B}). \quad (11)$$

As a result (9) simplifies to

$$C^F(\mathcal{B}) = \left( w \frac{1}{\theta} + p \right) \mathcal{B} + p \varphi \bar{\gamma} \left( \frac{\mathcal{B}}{s_0} \right)^{\frac{\gamma}{\gamma-1}} \quad (12)$$

where  $\bar{\gamma} \equiv \left(\frac{\gamma-1}{\gamma}\right)^\gamma$ .

Note that  $\gamma > 1$ . Hence, constructing infrastructure domestically has constant marginal costs, while using foreign engineers has *increasing* marginal costs. The reason is that the average fixed costs (per project size!) increase, because projects are getting smaller. Specifically, total fixed costs per quantity imported are given by

$$\frac{\text{Fixed costs}}{\mathcal{B}} = \frac{\left(\frac{s_0}{s^F(\mathcal{B})}\right)^\gamma p\varphi}{\mathcal{B}} = \frac{\frac{\gamma-1}{\gamma} p\varphi}{s^F(\mathcal{B})} = p\varphi \left(\frac{\gamma-1}{\gamma}\right)^{\frac{\gamma}{\gamma-1}} s_0^{-\frac{\gamma}{\gamma-1}} \mathcal{B}^{\frac{1}{\gamma-1}}, \quad (13)$$

i.e. total fixed costs per unit of infrastructure built are increasing in total construction with an elasticity  $\frac{1}{\gamma-1}$ . As the heterogeneity in project size disappears,  $\gamma \rightarrow \infty$ ,  $\frac{1}{\gamma-1} \rightarrow 0$  and this effect disappears. Hence, the *dispersion* in project sizes plays a crucial role for the comparative advantage of foreign engineers. Note also that this is the cost function as a function of investment *flow*. Hence, this is akin to an adjustment cost, given the fixed supply of a unit measure of investment projects.

In Figure 10 we show the two cost functions  $C^D(\mathcal{B})$  and  $C^F(\mathcal{B})$  as a function of  $\mathcal{B}$ . These are drawn for the case where foreign-run projects are cheaper on the margin, i.e.

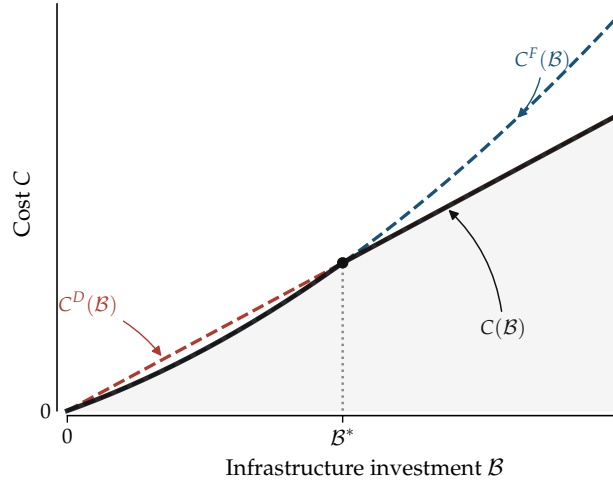
$$p < \frac{w}{q}. \quad (14)$$

Figure 10 implies a clear “pecking order”: the first units of infrastructure are large projects that are organized by foreign engineers. As infrastructure investment increases, projects are becoming smaller and fixed costs become more important. Eventually, foreign engineers lose their comparative advantage and the economy switches to domestic engineers.

Proposition 1 summarizes this discussion and derives the overall cost function; the envelope of the two cost functions shown in Figure 10.

**Proposition 1.** *Suppose that  $p < \frac{w}{q}$ , i.e. foreign engineers are cheaper on the margin. The*

Figure 10: The Infrastructure Cost Function



Notes: This figure shows the cost function  $C(B)$  in (15). The international (domestic) cost function is shown blue (red). The overall cost function is the lower envelope, shown in black.  $B^*$  (see (16)) denotes the level of construction when the country starts using domestic engineers to produce.

overall cost function of construction is given by

$$C(B) = \begin{cases} C^F(B) & \text{if } B \leq B^* \\ C^F(B^*) + C^D(B - B^*) & \text{if } B > B^* \end{cases}, \quad (15)$$

where

$$B^* = \left( \frac{\frac{w}{q} - p}{p\varphi\bar{\gamma}} \right)^{\gamma-1} s_0^\gamma. \quad (16)$$

Hence, for  $B < B^*$ , only foreign engineers are used. Once  $B > B^*$  domestic engineers are employed. The average project size is declining in  $B$ . If  $p > \frac{w}{q}$ , only domestic engineers are used,  $C(B) = C^D(B)$ , and average project size is not determined.

This implies also that the import share takes a simple form

$$\chi(B) = \min \left\{ 1, \frac{B^*}{B} \right\} \quad (17)$$

Holding  $B^*$  constant, an increase in infrastructure spending reduces the import

share because the economy reduces project size. But  $w$ ,  $q$ , and  $p$  affect both  $\mathcal{B}^*$  and  $\mathcal{B}$ .

## 5.2. Optimal Infrastructure Spending

Infrastructure is a factor of production according to (1), but provided by the government. We assume that a benevolent government chooses the total amount construction spending, paid for by lump-sum taxes, to maximize utility of the representative household. As we show in Appendix Section I.2, we can express the government's problem directly as a choice of foreign and domestic infrastructure  $\mathcal{B}^D$  and  $\mathcal{B}^F$ :

$$\max_{(\mathcal{B}^F, \mathcal{B}^D)} \left( \lambda (A\ell_P)^{\frac{\sigma-1}{\sigma}} + (1-\lambda) I^{\frac{\sigma-1}{\sigma}} \right)^{\frac{\sigma}{\sigma-1}} - px(\mathcal{B}^F) \quad (18)$$

where

$$\ell_P = L - \left( \frac{1}{\theta} + \frac{1}{q} \right) \mathcal{B}^D - \frac{1}{\theta} \mathcal{B}^F \quad (19)$$

$$I = I_0 + \mathcal{B}^D + \mathcal{B}^F \quad (20)$$

$$x(\mathcal{B}^F) = \mathcal{B}^F + \varphi \bar{\gamma} \left( \frac{\mathcal{B}^F}{s_0} \right)^{\frac{\gamma}{\gamma-1}}, \quad (21)$$

with  $\bar{\gamma} \equiv \left( \frac{\gamma-1}{\gamma} \right)^\gamma$ . The maximization problem in (18) captures the main tradeoff of infrastructure building. The government aims to maximize overall domestic consumption. Importing foreign engineer has a resource costs of  $px(\mathcal{B}^F)$ , where  $x(\mathcal{B}^F)$  is given in (21) and reflect the increasing cost of foreign sourcing though fixed costs. The benefits of foreign sourcing are that it saves on domestic engineers, which can then either work as production workers or in the construction sector under the guidance of foreign engineers; see (19). The solution to this problem determines overall construction spending as a function of parameters.

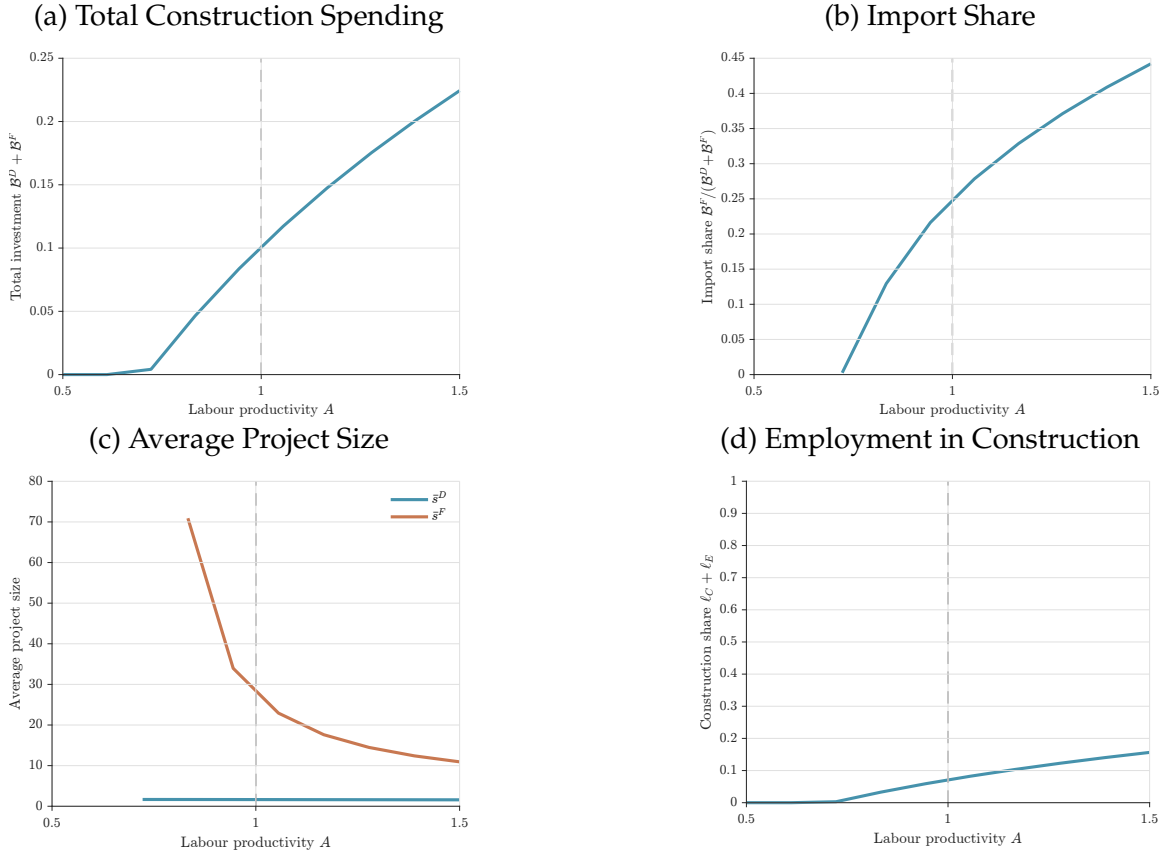
### 5.3. Characterizing the Solution

In this section we characterize the qualitative properties of the solution. The entire analysis uses the parameter values that we calibrate in Section 6. We focus on the two main determinants at the country-level: labor productivity  $A$  and the efficiency of domestic engineers  $q$ . For all cases, we display four outcomes: total construction demand  $\mathcal{B} = \mathcal{B}^F + \mathcal{B}^D$ , the import share  $\mathcal{B}^F/\mathcal{B}$ , the average size of imported and domestic projects, and the labor share of construction workers.

**The Effect of Labor Productivity  $A$**  Consider first an experiment where we vary labor productivity  $A$ , holding all other parameters constant. Variation in labor productivity  $A$  maps, of course, one-to-one to overall income  $Y$ . The results are contained in Figure 11. Higher labor productivity raises infrastructure spending, both because of production complementarities (with  $\sigma < 1$ , a higher  $A$  raises the marginal product of infrastructure) and because the relative price of foreign engineers declines. For very low levels of productivity, no new construction is built because the marginal product of construction is too low given the stock of infrastructure  $I_0$  in place. As seen in Panel (b), higher productivity also raises the share of construction that is sourced internationally. In Panel (c) we show the average project size. As expected, imported projects are bigger. Moreover, the larger the quantity of imported engineers, the smaller the average project size. Finally, in Panel (d), we show the share of people engaged in construction:  $\ell_C + \ell_E$ . In our model, this share is modestly *increasing* in the productivity of production workers  $A$ . This is due to the Baumol-type effect alluded to above: a higher  $A$  increases the marginal product of infrastructural investments.

**Domestic Engineering Capabilities  $q$**  In Figure 12 we display an experiment where we vary the efficiency of domestic engineers  $q$ . As shown in Panels (a) and (b), an increase in  $q$  increases overall construction demand but reduces the reliance

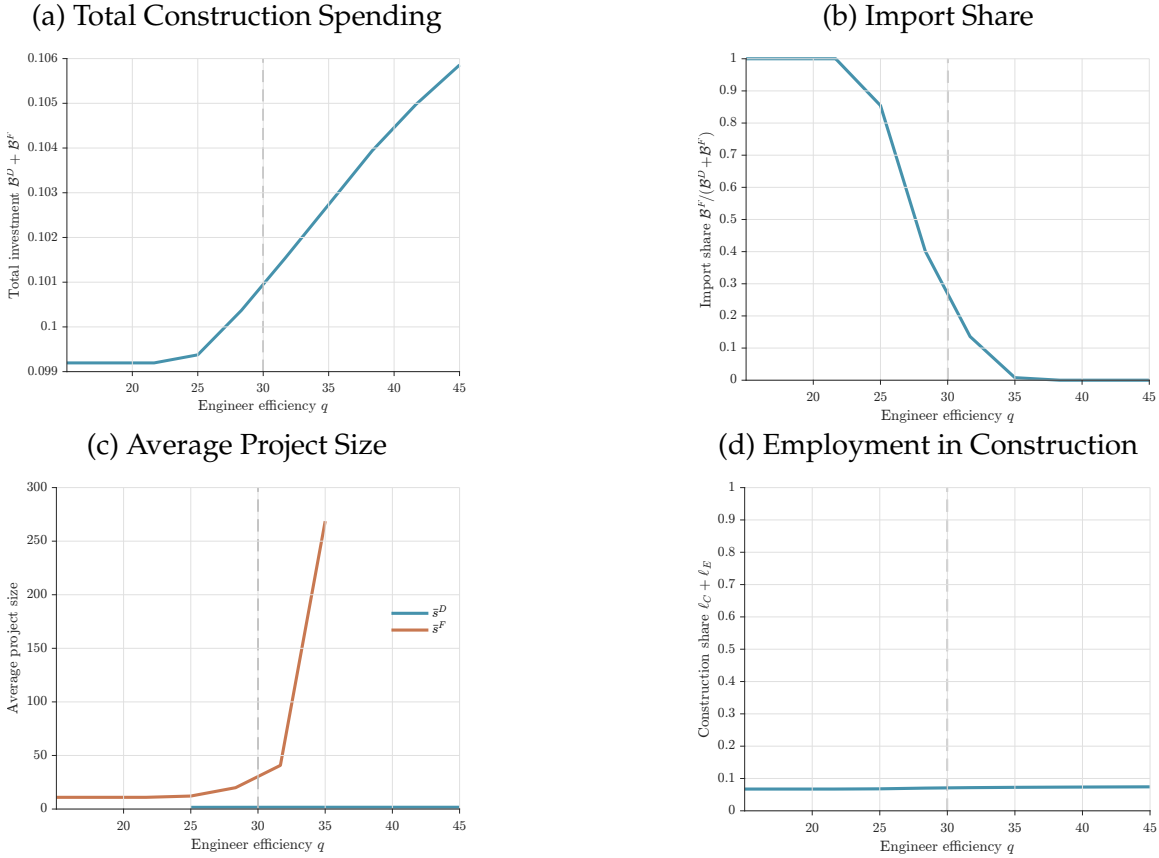
Figure 11: The Effect of Labor Productivity  $A$



Notes: The figure shows total construction spending  $B = B^F + B^D$ , the import share  $B^F/B$ , average project sizes  $\bar{s}^F$  and  $\bar{s}^D$ , and the employment share in construction activities,  $\ell_C + \ell_E$  as a function of productivity  $A$ . All other parameters are held constant. The initial parameters stem from the calibration discussed in Section 6.

on foreign engineers. The decline in foreign engineers also increases the average size of imported projects, because less and less projects are actually completed with foreign expertise. Finally, the overall employment share in construction-related activities is roughly constant as a function of engineering expertise  $q$ .

Figure 12: The Effect of Engineer Efficiency  $q$



Notes: The figure shows total construction spending  $B = B^F + B^D$ , the import share  $B^F/B$ , average project sizes  $\bar{s}^F$  and  $\bar{s}^D$ , and the employment share in construction activities,  $\ell_C + \ell_E$  as a function of engineering capability  $q$ . All other parameters are held constant. The initial parameters stem from the calibration discussed in Section 6.

## 6. Quantitative Analysis

### 6.1. Calibration Strategy

We calibrate our model in the following way. We assume that countries differ in two dimensions: labor productivity  $A_c$  and domestic engineering capabilities

Table 2: Calibration Strategy and Structural Parameters

|           | <i>Parameter</i>                    | <i>Value</i> | <i>Target Moment</i>                     |
|-----------|-------------------------------------|--------------|------------------------------------------|
| $A$       | Labor productivity                  | $\{A_c\}$    | GDP pc in country $c$                    |
| $q$       | Domestic engineering capability     | $\{q_c\}$    | Import share in country $c$              |
| $p$       | World price of engineering services | 0.016        | US New flow of construction / GDP        |
| $\theta$  | Construction labor productivity     | 1.47         | US Construction empl. share ( $\ell_P$ ) |
| $\varphi$ | Fixed cost of foreign engineers     | 4.63         | Share of fixed costs in foreign projects |
| $I_0$     | Initial stock of infrastructure     | 1            | Stock of infrastructure / GDP            |
| $\sigma$  | Elasticity of substitution          | 0.5          | Exogenous                                |
| $\lambda$ | CES Weight on labor                 | 0.5          | Normalization                            |
| $\gamma$  | Shape of project size distribution  | 2.5          | Exogenous                                |
| $L$       | Labor endowment                     | 1            | Normalization                            |
| $s_0$     | Minimum project size                | 1            | Normalization                            |

Notes: This table reports the structural parameters, their calibrated parameters and the mapping to the targeted moments.

$q_c$ . All the other parameters are common across countries. We therefore calibrate our model to moments from a typical rich country, e.g. the US, and then vary  $(A_c, q_c, I_0)$  to perfectly rationalize the joint distribution of income per capita and import shares shown in Figure 8, while anchoring the initial stock of infrastructure  $I_0$  to the level of GDP.

In Table 2 we summarize the structural parameters. Even though we calibrate all parameters jointly, we indicate the main target moment that is informative about the parameter of interest. That  $A$  and  $q$  are informative about GDP pc and the import share is obvious given that  $q$  parametrizes the comparative advantage of domestic engineering inputs. The productivity of construction workers  $\theta$  is a key driver of the employment share in construction. Separately identifying  $p$  and  $\varphi$  is challenging because both directly affect the cost of foreign engineers. We calibrate  $\varphi$  to match a target for the share of fixed costs in the provision of internationally-sourced projects (see (13)) and  $p$  to match a given target of the level of construction output.

Table 3: Model Fit

| Parameter | Description                        | Value   | Moment                                        | Target | Model  |
|-----------|------------------------------------|---------|-----------------------------------------------|--------|--------|
| $\theta$  | Construction productivity          | 1.4735  | $\ell_P/L$                                    | 90%    | 92.9%* |
| $q$       | Domestic engineer efficiency       | 30.00   | $\mathcal{B}^F/(\mathcal{B}^D+\mathcal{B}^F)$ | 25%    | 25.0%  |
| $\varphi$ | Fixed-cost scale                   | 4.6312  | $FC/x(\mathcal{B}^F)$                         | 10%    | 10.0%  |
| $p$       | World price of foreign engineering | 0.01655 | $(\mathcal{B}^D+\mathcal{B}^F)/Y$             | 10%    | 10.0%  |

Notes: The table reports the calibrated parameters and the respective moments. The other parameters are  $I_0 = 1$ ,  $\sigma = 0.5$ ,  $\lambda = 0.5$ ,  $\gamma = 2.5$ ,  $L = 1$ ,  $s_0 = 1$ ,  $A = 1$ . \* The interior FOC implies  $\ell_P/I = \sqrt{1/\theta + 1/q}$  (structural ceiling  $\approx 93.4\%$ ); moments (2)–(4) match within 0.2%.

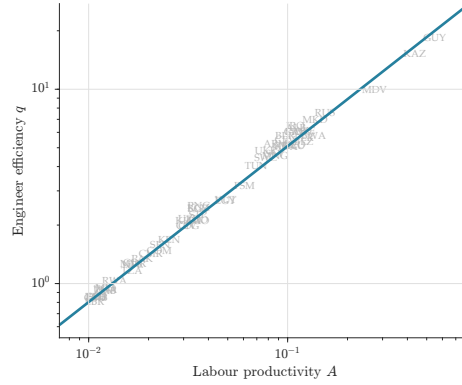
## 6.2. Results and Model Fit

In Table 3 we report the fit of the model for the US. As seen in the table, we target an employment share of production workers of 90%, an import share of 25% and an overall flow of construction investment of 10% of GDP. We also discipline the fixed cost parameter  $\varphi$  so that 10% of the entirety of foreign engineering imports are “soaked up” to pay for fixed costs. Table 3 shows that we can match these moments relatively well.<sup>5</sup>

In Figure 13 we report the calibrated tuples of  $\{A_c, q_c\}$  to match the joint distribution of GDP pc and import shares reported in Figure 8. To rationalize the data on income per capita and countries’ reliance on foreign engineering services, our theory points towards a positive correlation between overall productivity  $A_c$  and engineering capabilities  $q_c$ .

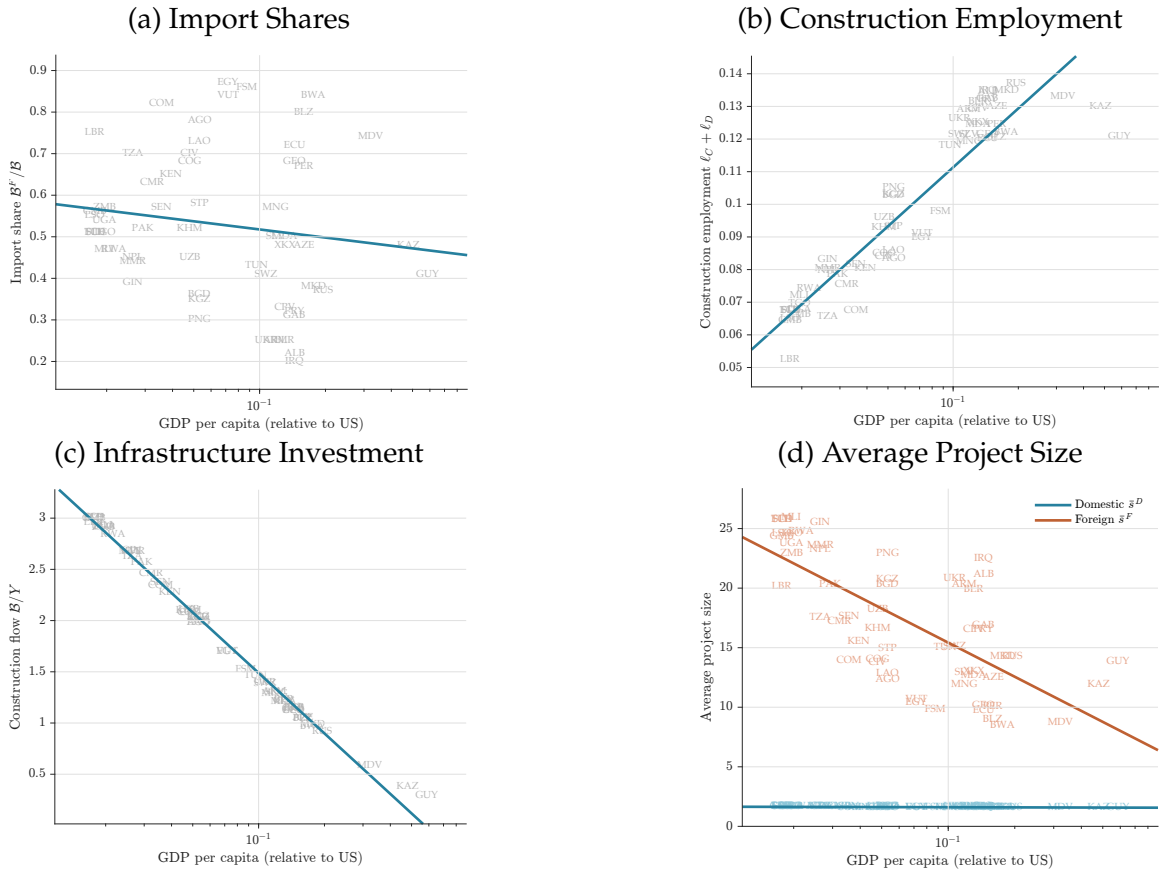
<sup>5</sup>We slightly overestimate the share of production worker, which is due to our exogenous labor share in the production function  $\lambda$ .

Figure 13: Labor Productivity and Engineering Capabilities



Notes: This figure plots the combination of productivity  $A$  and engineering capability  $q$  that rationalizes the data on import shares and GDP per capita from Figure 8.

Figure 14: Cross country variation in construction



Notes: This figures reports the cross-country correlation between GDP pc and import shares (Panel (a)), employment in construction activities, i.e.  $\ell_C + \ell_E$  (Panel (b)), construction investment relative to BGP (Panel (c)), and the average size of foreign and domestic projects (Panel (d)).

## 7. Conclusion

TBD

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# Appendix

## H. Data

Table A.1: Availability of ENR Reports by Year

| Year | Top List | Where Worked | Year | Top List | Where Worked |
|------|----------|--------------|------|----------|--------------|
| 1989 | ✓        | ✓            | 2008 | ✓        | ✓            |
| 1990 | ✓        | ✓            | 2009 | ✓        | ✓            |
| 1991 | ✓        | ✓            | 2010 | ✓        | ✓            |
| 1992 | ✓        | ✓            | 2011 | ✓        | ✓            |
| 1993 | ✓        | ✓            | 2012 | ✓        | ✓            |
| 1994 | ✓        | ✓            | 2013 | ✓        | ✓            |
| 1995 | ✓        | ✓            | 2014 | ✓        | ✓            |
| 1996 |          | ✓*           | 2015 | ✓        | ✓            |
| 1997 |          | ✓*           | 2016 | ✓        | ✓            |
| 1998 |          | ✓*           | 2017 | ✓        | ✓            |
| 1999 |          |              | 2018 | ✓        | ✓            |
| 2000 |          |              | 2019 | ✓        | ✓            |
| 2001 | ✓        | ✓            | 2020 | ✓        | ✓            |
| 2002 | ✓        |              | 2021 | ✓        | ✓            |
| 2003 | ✓        |              | 2022 | ✓        | ✓            |
| 2004 | ✓        |              | 2023 | ✓        | ✓            |
| 2005 | ✓        | ✓            | 2024 | ✓        | ✓            |
| 2006 | ✓        | ✓            | 2025 | ✓        | ✓            |
| 2007 | ✓        | ✓            |      |          |              |

*Note:* Years for which the ENR top list and the “where worked” supplement are available.

\* no top list available, but the supplement is available. Data in each report refer to the prior calendar year.

## I. Theory: Derivations

### I.1. Project Size

A key implication of our theory is that the average project size differs between imported and domestic projects. For foreign projects, the average size is given by

$$\bar{s}^F = E[s | s \geq s^F(\mathcal{B}^F)] = \frac{\gamma}{\gamma - 1} s^F(\mathcal{B}^F). \quad (\text{A.1})$$

By comparison the average size of domestic project, assuming that a random sample of the remaining projects are drawn, is given by

$$\bar{s}^D = E[s | s < s^F(\mathcal{B}^F)] = E[s | s < s^F] = \frac{\gamma}{\gamma - 1} \cdot \frac{s_0 - s_0 \left( \frac{s_0}{s(\mathcal{B}^F)} \right)^{\gamma-1}}{1 - \left( \frac{s_0}{s(\mathcal{B}^F)} \right)^{\gamma}} \quad (\text{A.2})$$

### I.2. Derivation of Government's Problem

We can posit the problem of the government using the primal approach, where the government chooses all employment allocations  $(\ell_P, \ell_E, \ell_C)$  and the amount of foreign engineering services  $x$  to maximize domestic consumption.

Under our assumptions, engineers and construction are always used in strict proportions. Per unit of size, one needs  $\theta^{-1}$  units of construction labor and one unit of engineering human capital. Hence, the total amount of production workers required is

$$\ell_C = \frac{1}{\theta} (\mathcal{B}^D + \mathcal{B}^F). \quad (\text{A.3})$$

As for engineers, one requires

$$\ell_E = \frac{1}{q} \mathcal{B}^D \quad (\text{A.4})$$

domestic engineers. And to build  $\mathcal{B}^F$  units of foreign infrastructure one requires

$x(\mathcal{B})^F$  units of foreign engineers, which are given by (see (12))

$$x(\mathcal{B}^F) = \mathcal{B}^F + \varphi \bar{\gamma} \left( \frac{\mathcal{B}^F}{s_0} \right)^{\frac{\gamma}{\gamma-1}} \quad (\text{A.5})$$

where  $\bar{\gamma} \equiv \left( \frac{\gamma-1}{\gamma} \right)^\gamma$ .

Using the labor market clearing condition

$$L = \ell_P + \ell_C + \ell_D = \ell_P + \frac{1}{\theta} (\mathcal{B}^D + \mathcal{B}^F) + \frac{1}{q} \mathcal{B}^D, \quad (\text{A.6})$$

we can express the government's problem as a simple choice of  $\mathcal{B}^D$  and  $\mathcal{B}^F$ :

$$\max_{(\mathcal{B}^F, \mathcal{B}^D)} \left( \lambda (A \ell_P)^{\frac{\sigma-1}{\sigma}} + (1-\lambda) I^{\frac{\sigma-1}{\sigma}} \right)^{\frac{\sigma}{\sigma-1}} - p x(\mathcal{B}^F) \quad (\text{A.7})$$

where

$$\ell_P = L - \left( \frac{1}{\theta} + \frac{1}{q} \right) \mathcal{B}^D - \frac{1}{\theta} \mathcal{B}^F \quad (\text{A.8})$$

$$I = I_0 + \mathcal{B}^D + \mathcal{B}^F \quad (\text{A.9})$$

$$x(\mathcal{B}^F) = \mathcal{B}^F + \varphi \bar{\gamma} \left( \frac{\mathcal{B}^F}{s_0} \right)^{\frac{\gamma}{\gamma-1}}, \quad (\text{A.10})$$

with  $\bar{\gamma} \equiv \left( \frac{\gamma-1}{\gamma} \right)^\gamma$

## J. Theory: Derivations

### J.1. Derivation of the planning problem (??)

The households' budget constraint is  $c_t L = w_t L + T_t$ . Substituting the government budget constraint (??) gives

$$c_t L = w_t L - w_t(\ell_{Et} + \ell_{Ct}) - pS_t = w_t(L - \ell_{Et} - \ell_{Ct}) - pS_t.$$

The wage equals the marginal product of labor in the final-goods sector:

$$w_t = \frac{\partial f(A(L - \ell_{Et} - \ell_{Ct}), I_t)}{\partial \ell_{tP}}.$$

Substituting these two expressions into the household's utility function yields the planning problem (??).

### J.2. Derivation of the special case (??)

Taking  $N$  to be continuous and solving the government's optimal allocation of construction workers across traditional and modern projects, the total construction flow is

$$\mathcal{B}(\ell_{Ct}, q\ell_{Et}, S_t) = \ell_{Ct}^\eta \left( 1 + \left( \frac{Z}{F^{1-\eta}} \right)^{\frac{1}{1-\eta}} (q\ell_{Et} + S_t) \right)^{1-\eta}.$$

The derivation proceeds by noting that at an interior optimum, the government equates the marginal product of a construction worker in the traditional technology to that in a modern project. Since all active modern projects are identical in their use of construction workers (as fixed costs are in engineering services only), this pins down the optimal size of each project and the optimal number of projects, which together yield the aggregate expression above.

### J.3. Do we need a functional form for $H$ ?

The analysis separates cleanly into a part that is distribution-free and a part that is not. The *sign* of the policy result—that a small tax on foreign engineers raises welfare under the learning externality—holds for any size distribution with positive density at the cutoff: by (A.20) the marginal effect depends on  $H$  only locally, through  $h(s_0^*)$ . The same is true of the qualitative comparative statics in  $q_1$ ,  $A$ , and  $p$ .

The *magnitudes*—the level of the optimal tax, the size of the welfare gains, and the full cross-country profile—are integrals of project size over the region between the free-trade and protected cutoffs and therefore depend on the global shape of  $H$ . A Pareto distribution  $H(s) = 1 - (s_{\min}/s)^k$  makes these closed-form: the import share is

$$\chi(s^*) = \left(\frac{s_{\min}}{s^*}\right)^{k-1}, \quad E_D = T[1 - \chi(s^*)], \quad T = \frac{k s_{\min}}{k-1}, \quad (\text{A.11})$$

which, combined with (A.21), delivers an explicit optimal tax. Alternatively, and preferably for quantification,  $H$  can be taken directly from the empirical distribution of works-contract values in the World Bank procurement data, which is the empirical counterpart of  $H$ . In short: no functional form is needed to establish that the tax helps and to sign its cross-country pattern; a functional form—Pareto for analytics, or the empirical distribution for the calibration—is needed only to compute magnitudes.

## K. The Distribution of Construction Contract Sizes

The model delivers sharp predictions about the size distribution of construction projects, and these predictions differ depending on whether project sizes are lognormal or instead follow a Pareto (power-law) tail. This appendix asks which of the two families is the better empirical description. The data are the universe of World Bank major contract awards over 2000–2025, comprising 102,118 contracts

with a positive value, which we convert from nominal to constant 2020 US dollars using the U.S. CPI. We emphasize that the goal is to gauge *relative* fit rather than to conduct a formal goodness-of-fit test of either functional form.

We fit the lognormal by maximum likelihood over the full positive support and the Pareto by maximum likelihood to the upper tail above a cutoff  $x_{\min}$ , and we compare the two using the normalized log-likelihood-ratio statistic of ?, in which a positive value indicates that the lognormal is the closer fit. Because the appropriate tail cutoff is itself uncertain, we report the comparison across a range of cutoffs from \$1,000 to \$10 million.

The lognormal is decisively closer to the data at every cutoff (Table A.2). The estimated maximum-likelihood lognormal has  $\hat{\mu} = 11.82$  and  $\hat{\sigma} = 2.64$  (a median contract of roughly \$135,000), and its Kolmogorov–Smirnov distance from the empirical distribution never exceeds 0.05, an order of magnitude smaller than that of the fitted Pareto. Figure A.1 makes the comparison visually. In the left panel, the empirical complementary CDF (plotted on log–log axes) is tracked by the lognormal across all nine orders of magnitude of contract value, whereas the Pareto fitted to the tail above \$100,000 is too heavy: beyond roughly \$10 million the data fall *below* the power law, so large contracts are rarer than a Pareto would imply. The right panel shows that log contract values line up closely with lognormal quantiles, with only mild departures in the extreme tails.

Three features make this conclusion robust. First, the verdict is unchanged whether amounts are measured in nominal or constant dollars. Second, it does not depend on the many small contracts: excluding awards below \$10,000 (a fifth of the sample, plausibly executed with traditional, non-engineered technologies) leaves the comparison essentially unchanged, consistent with a single lognormal generating the entire size range rather than a distinct small-project regime. Third, the result is not an artifact of pooling heterogeneous recipients: estimating the comparison separately for each of the twelve largest borrowing countries favors

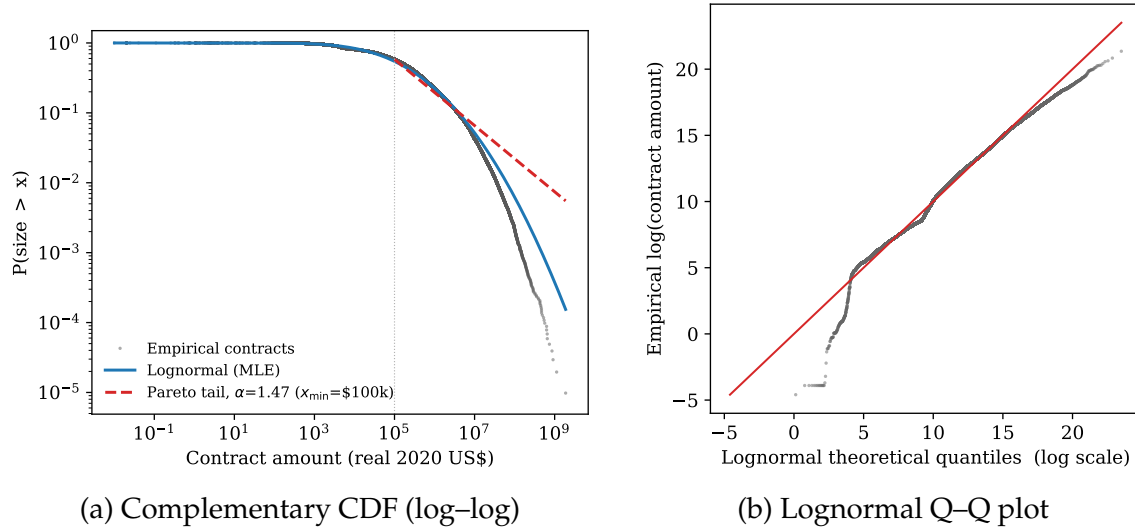
Table A.2: Lognormal Versus Pareto: Relative Fit of the Contract Size Distribution

| Tail cutoff $x_{\min}$ | $N \geq x_{\min}$ | $\hat{\alpha}$<br>(Pareto) | $D_{\text{lognormal}}$<br>(KS) | $D_{\text{Pareto}}$<br>(KS) | LR<br>(ln vs. Pareto) |
|------------------------|-------------------|----------------------------|--------------------------------|-----------------------------|-----------------------|
| \$1,000                | 99,292            | 1.20                       | 0.050                          | 0.242                       | +155.7                |
| \$10,000               | 82,334            | 1.28                       | 0.012                          | 0.204                       | +130.1                |
| \$100,000              | 59,442            | 1.47                       | 0.011                          | 0.106                       | +57.1                 |
| \$1,000,000            | 22,795            | 1.72                       | 0.014                          | 0.069                       | +23.9                 |
| \$10,000,000           | 4,346             | 2.17                       | 0.013                          | 0.038                       | +6.1                  |

*Notes.* World Bank major contract awards, 2000–2025, in constant 2020 US dollars ( $N = 102,118$  with positive value). For each tail cutoff  $x_{\min}$ ,  $\hat{\alpha}$  is the maximum-likelihood Pareto exponent on  $x \geq x_{\min}$ ;  $D$  is the Kolmogorov–Smirnov distance of each fitted distribution from the empirical CDF (smaller is better). LR is the normalized log-likelihood-ratio statistic of ? comparing the lognormal to the Pareto, with positive values indicating that the lognormal is the closer fit; all five are significant at the 1% level. The fitted lognormal over the full support has  $\hat{\mu} = 11.82$ ,  $\hat{\sigma} = 2.64$ .

the lognormal in every case. A final diagnostic reinforces the point. The Pareto tail exponent is not stable but rises monotonically with the cutoff, from  $\hat{\alpha} \approx 1.2$  at \$1,000 to  $\hat{\alpha} \approx 2.2$  at \$10 million. A genuine power law has a cutoff-invariant exponent; a drifting exponent of this kind is the signature of an underlying lognormal (?). We therefore treat project sizes as lognormal in the quantitative analysis, while noting that the relative advantage of the lognormal narrows in the extreme upper tail, where the two families are hardest to distinguish.

Figure A.1: Contract Sizes Are Lognormal, Not Pareto



*Notes.* World Bank major contract awards, 2000–2025, in constant 2020 US dollars. Panel (a) plots the empirical complementary CDF (grey) with the maximum-likelihood lognormal (solid) and a Pareto fitted to the tail above \$100,000 (dashed); the dotted vertical line marks the \$100,000 cutoff. Panel (b) plots empirical log contract values against lognormal theoretical quantiles; the line is the 45° reference.

## L. Additional Empirical Figures

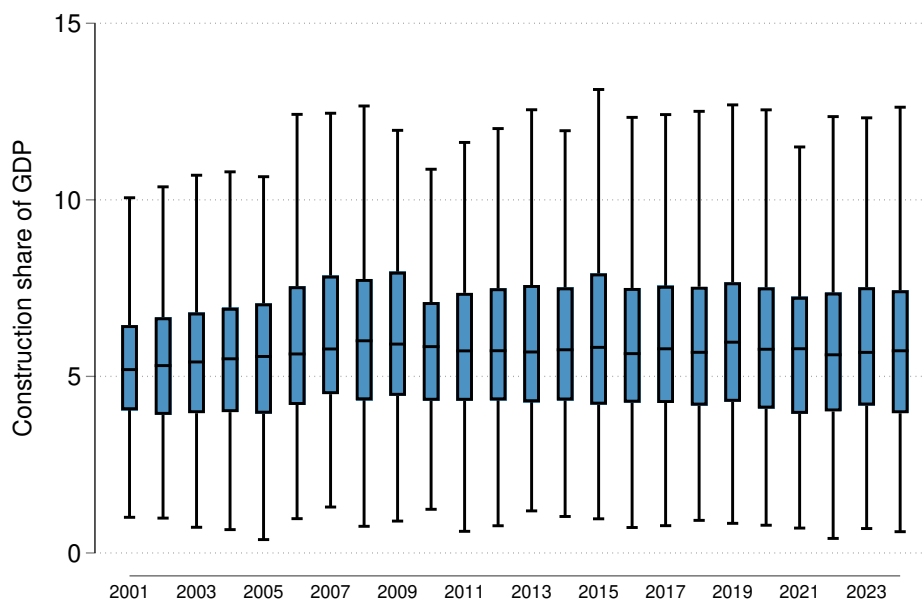
## M. OLD Theory section

Table A.3: Official Direct-Investment Positions in Construction

|                                                            | United States<br>(2024) | China<br>(2024) | EU (extra-EU)<br>(2023) |
|------------------------------------------------------------|-------------------------|-----------------|-------------------------|
| <i>Official outward direct investment (stock/position)</i> |                         |                 |                         |
| Total outward FDI position <sup>a</sup>                    | 6,830                   | 3,140           | 9,036 <sup>†</sup>      |
| of which: Construction <sup>b</sup>                        | 8                       | 56              | 78 <sup>†</sup>         |
| Construction share of total                                | < 1%                    | < 2%            | < 1%                    |

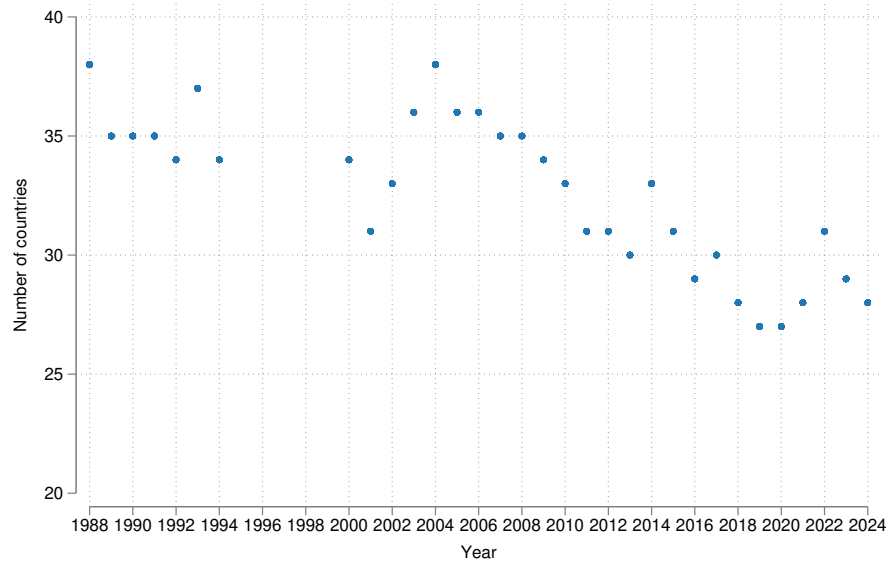
*Notes.* Values in current US\$ billion unless marked <sup>†</sup> (EUR billion). US: BEA, U.S. direct investment abroad, historical-cost position, end-2024. China: MOF-COM/SAFE, ODI stock, end-2024. EU: Eurostat `bop_fdi6_pos`, extra-EU outward position, end-2023.

Figure A.2: Size of the Construction Sector



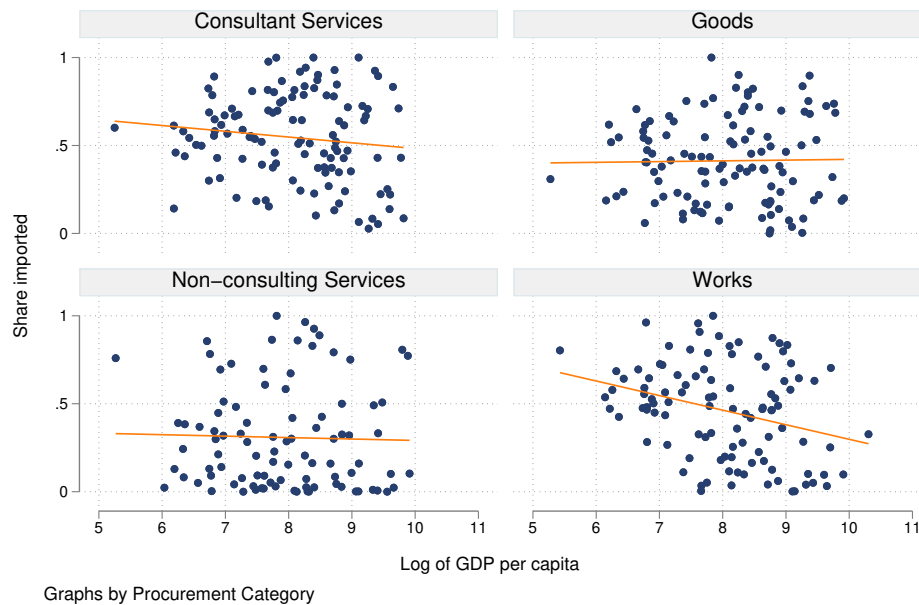
*Note:* Box-and-whisker plot of the construction sector's share in GDP across countries, 2001–2024. The line inside each box is the median; the box spans the interquartile range. Source: UN National Accounts.

Figure A.3: Number of Source Countries for Top Global Contractors



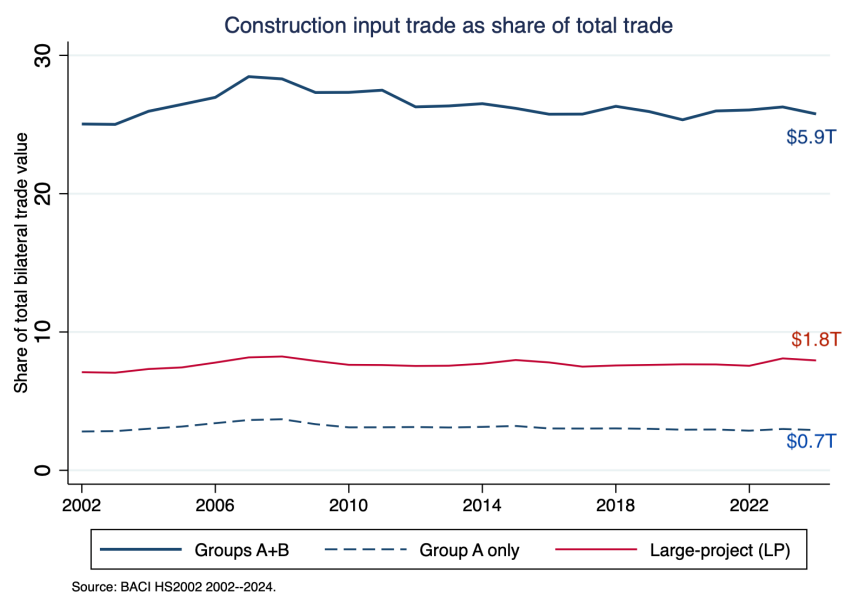
*Note:* Number of countries from which any of the top-250 firms originate, by year. Source: ENR.

Figure A.4: Import Share by Procurement Category



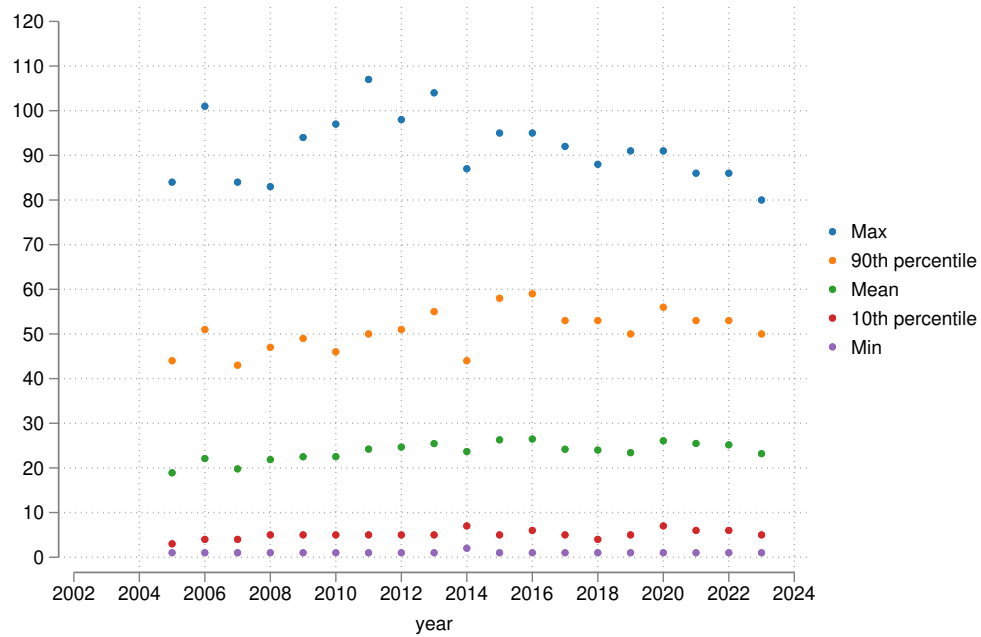
*Note:* Share of contract value provided by foreign-registered suppliers by procurement category and borrower income group, 2020–2026. Works account for 56% of total contract value, goods for 25%, consultant services for 13%, and non-consulting services for 6%. Source: World Bank IPF.

Figure A.5: Construction Input Trade as a Share of Total Merchandise Trade



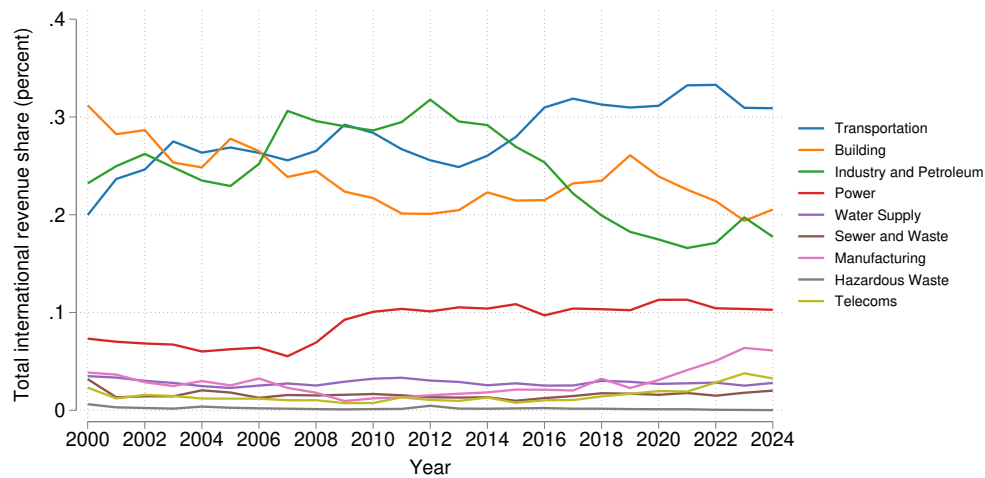
*Note:* Share of global merchandise trade accounted for by each construction-input category, 2002–2024. LP = large-project indicator.  
Source: BACI CEPII.

Figure A.6: Number of Foreign Firms Working in Each Country, 2005–2023



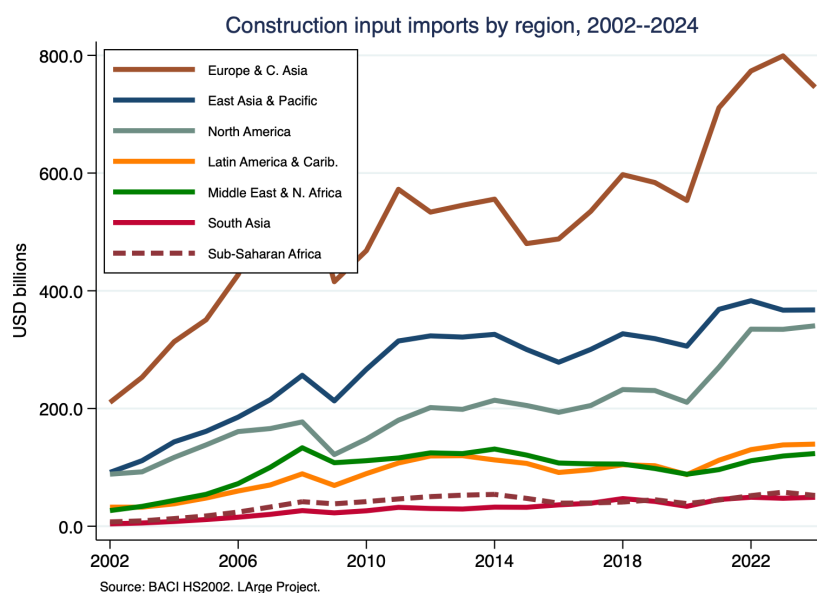
Note: Number of top-250 firms working in each country in each year 2005–2023. Source: ENR.

Figure A.7: Sectors of Operation of the Top Firms



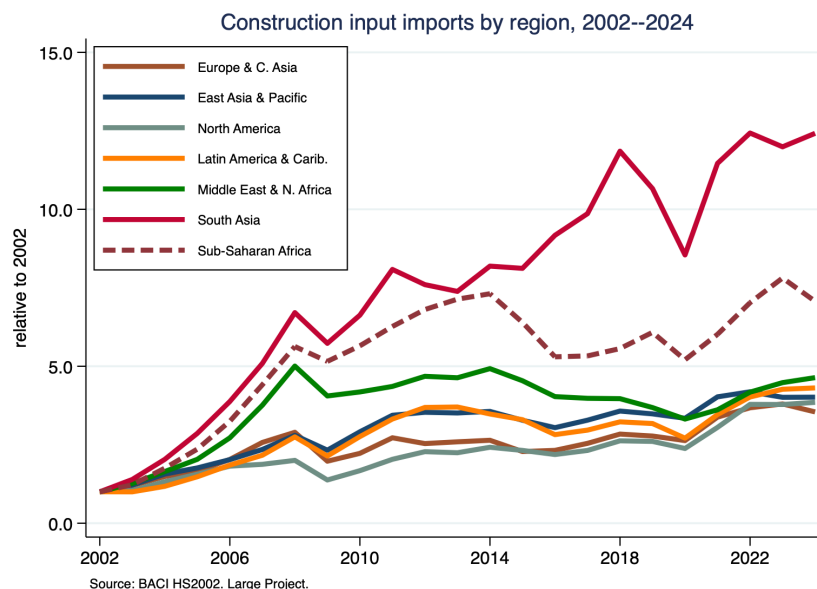
Note: Share of top-250 firms active in each sector. Firms can operate in multiple sectors. Source: ENR.

Figure A.8: Construction Input Imports by Region: Levels



*Note:* Value of large-project construction input imports by World Bank region, 2002–2024. Source: BACI CEPII.

Figure A.9: Construction Input Imports by Region: Growth (2002 = 100)



*Note:* Large-project construction input imports indexed to 2002 = 100, by World Bank region. South Asia and Sub-Saharan Africa are the fastest-growing regions. Source: BACI CEPII.

Figure A.10: China's Export Share: Construction Inputs vs. Total Merchandise



*Note:* China's share of world exports separately for large-project construction inputs (LP) and total merchandise, 2002-2024. Source: BACI CEPII.

**Government.**

$$c = Y - \int_{\mathcal{N}_F} (ps + \varphi) dH(s), \quad (\text{A.12})$$

where  $\mathcal{N}_F \subseteq \mathcal{N}$  is the set of imported projects. Labor is allocated across production, construction, and domestic engineering. Writing  $\ell_E = \frac{1}{q} \int_{\mathcal{N}_D} s dH(s)$  for domestic engineering labor with  $\mathcal{N}_D = \mathcal{N} \setminus \mathcal{N}_F$  the domestically engineered projects, each requiring  $s/q$  engineers. The labor market clears as

$$\ell_P + \ell_C + \ell_E = L, \quad \ell_C = \theta I, \quad \ell_E = \frac{1}{q} \int_{\mathcal{N}_D} s dH(s), \quad (\text{A.13})$$

and the common wage equals the marginal product of production labor,  $w = \partial Y / \partial \ell_P$ . Construction labor  $\ell_C$  and domestic engineering labor  $\ell_E$  both crowd out production and so enter consumption through  $\ell_P$ ; imported engineering is instead paid abroad, which is why only it appears explicitly in (A.12).

### M.1. Sorting and the static equilibrium

**Which source: sorting by project size** Consider the choice of source for a project of size  $s$  that is worth completing. Domestic engineering costs  $\frac{w}{q}s$ ; importing costs  $ps + \varphi$ . The economy is in the empirically relevant regime when foreign engineering is cheaper at the margin but carries the fixed cost,  $p < w/q$ : large enough projects amortize  $\varphi$  and are cheaper to import, while small projects are cheaper to execute domestically.

**Proposition 2** (Sorting). *Suppose  $p < w/q$ . There is a unique size threshold*

$$s^* = \frac{\varphi}{w/q - p} \quad (\text{A.14})$$

*such that every completed project with  $s < s^*$  is engineered domestically and every completed project with  $s > s^*$  is imported. The threshold  $s^*$  is increasing in the foreign fixed cost  $\varphi$ , increasing in the domestic unit cost  $w/q$ , and decreasing in the world price  $p$ . If  $p \geq w/q$ , all engineering is imported.*

*Proof.* Importing project  $s$  is cheaper than executing it domestically iff  $ps + \varphi < (w/q)s$ , i.e.  $s(w/q - p) > \varphi$ . When  $w/q > p$  this holds iff  $s > s^*$  with  $s^*$  as in (A.14); the comparative statics follow by inspection. When  $w/q \leq p$  the inequality holds for all  $s$  (with  $\varphi \geq 0$ ), so all projects are imported.  $\square$

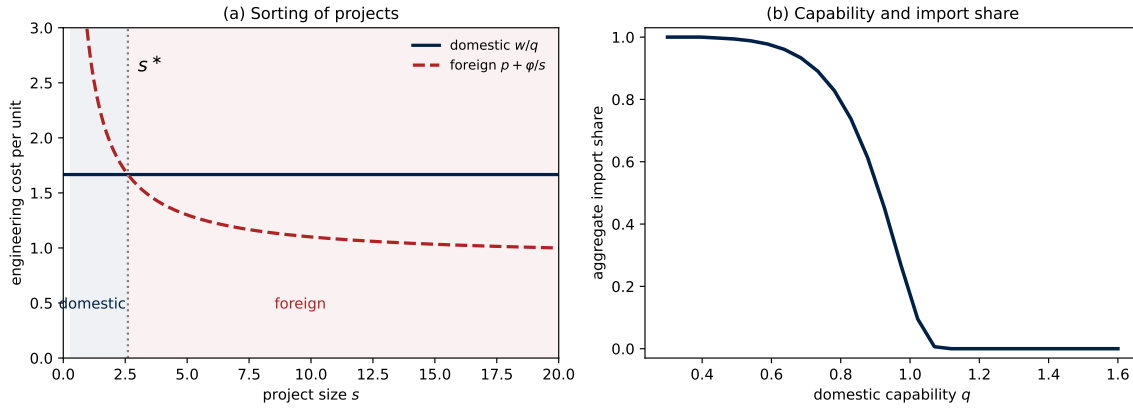
Sorting at the project level produces an interior allocation at the *country* level: domestic engineers and foreign contractors are used simultaneously, on different projects, exactly as in the data. The model reproduces the non-overlapping size distributions of imported and domestic contracts directly, with the crossover located at  $s^*$ . Panel (a) of Figure A.11 shows the two per-unit cost schedules and the threshold.

**Which projects: the extensive margin** A project is completed only if its contribution to output covers its cost. Completing project  $s$  raises output by  $m s$ , where  $m \equiv \partial Y / \partial I$  is the marginal product of infrastructure, and costs  $w\theta s$  in construction wages, paying its  $\theta s$  construction workers, which add to  $\ell_C$ , plus its engineering cost. Per unit of size, the surplus from a domestically engineered project is  $m - w\theta - w/q$  (constant in  $s$ ), and from an imported project is  $m - w\theta - p - \varphi/s$  (increasing in  $s$ ). Because the construction term  $w\theta$  is common to both sources, it does not affect the sourcing cutoff,  $s^*$ ; it does shape the build margin below.

**Lemma 1** (Build set). *Let  $m$  and  $w$  be the equilibrium marginal product of infrastructure and wage. If  $m - w\theta \geq w/q$ , every project is completed: those with  $s < s^*$  domestically and those with  $s > s^*$  by import. If  $p < m - w\theta < w/q$ , only large projects are completed and all are imported:  $\mathcal{N} = \{s : s \geq s_b\}$  with  $s_b = \varphi / (m - w\theta - p) > s^*$  and  $\mathcal{N}_D = \emptyset$ . If  $m - w\theta \leq p$ , no modern project is completed.*

The equilibrium is a fixed point: the build and sourcing cutoffs determine  $I$  and the labor allocation; these determine  $m$  and  $w$  through (1); and these in turn must reproduce the cutoffs. Existence follows from continuity and the monotonicity of  $m$  in  $I$ .

Figure A.11: The sorting mechanism.



Note: Panel (a): a project of size  $s$  is engineered domestically if  $s < s^*$  and imported if  $s > s^*$ , where  $s^* = \varphi/(w/q - p)$  equalizes the per-unit cost of the two sources. Panel (b): the aggregate import share as a function of domestic engineering capability  $q$ ; at low  $q$  the capacity constraint binds and the country imports everything. Computed from the static equilibrium of Section M.1 with lognormal project sizes.

**Definition 1** (Aggregate import share). *The aggregate import share is the value of imported engineering relative to all engineering,  $\chi = \frac{\int_{\mathcal{N}_F} s dH}{\int_{\mathcal{N}} s dH}$ .*

## M.2. Country heterogeneity

Countries differ in productivity  $A$  and engineering capability  $q$ . The model maps these primitives into the import share and matches the cross-country facts.

**Proposition 3** (Capability and the capacity constraint). *Holding  $w$  fixed, a lower domestic capability  $q$  raises the effective unit cost  $w/q$ , lowers the sorting threshold  $s^*$ , and raises the import share  $\chi$ . Below a capability cutoff  $\underline{q}$  defined by  $w/\underline{q} = p + \varphi/s_{\min}$ , all projects are imported ( $\chi = 1$ ).*

Proposition 3 is the model's account of Fact 3. The poorest countries have so little domestic engineering capability that  $w/q$  exceeds the per-unit import cost even for the smallest projects; they import everything and operate at the corner  $\chi = 1$ —the binding capacity constraint. As capability rises, the threshold  $s^*$  moves into the interior of the size distribution and the country begins to execute its smaller projects at home. Panel (b) of Figure A.11 traces  $\chi$  against  $q$  and shows the transition from

the all-import corner to interior sorting.

**Proposition 4** (Productivity and imports). *In general equilibrium a higher productivity  $A$  raises the wage  $w$  and hence the opportunity cost of domestic engineering  $w/q$ , lowering  $s^*$  and raising the import share. Productive economies import a larger share of their engineering because skilled labor is more valuable in production.*

### M.3. Project complexity and frontier expertise

The baseline makes a stark prediction at the top of the capability distribution: once  $w/q < p$ , domestic engineering is cheaper for every project and the import share is exactly zero. The data say otherwise. Capable, high-income countries continue to import specialized infrastructure—the United States alone hosts more than fifty foreign contractors. This section adds the minimal ingredient that captures this fact without disturbing any of the low-income results: a second dimension of project heterogeneity on which global firms hold an advantage that no level of  $q$  overturns.

Let each project carry, in addition to its size  $s$ , a complexity index  $z \in [0, 1]$ , distributed independently with density  $f(z)$ . Complexity captures proximity to the technological frontier: signature structures, deep tunnelling, LNG and nuclear works, or high-speed rail. Domestic engineers have efficiency  $q g(z)$  on a project of complexity  $z$ , where  $g$  is decreasing with  $g(0) = 1$ : local capability degrades on more complex, frontier projects. Global contractors operate at the world frontier and are complexity-neutral, supplying engineering at efficiency one and price  $p$ . This asymmetry is the formal counterpart of the non-rivalry of expertise emphasized throughout the paper. A global firm's accumulated know-how, built across dozens of comparable projects worldwide and redeployed at near-zero marginal cost, is precisely what a domestic sector that has executed a handful of projects cannot match on the most demanding work.

The domestic cost per efficiency unit at complexity  $z$  is increasing in complexity

$$a(z) \equiv \frac{w}{q g(z)}, \quad a'(z) > 0. \quad (\text{A.15})$$

Sourcing follows the same comparison as before, now project-by-project in  $(s, z)$  space: project  $(s, z)$  is imported iff  $ps + \varphi < a(z)s$ , i.e. iff

$$a(z) > p \quad \text{and} \quad s > s^*(z) = \frac{\varphi}{a(z) - p}. \quad (\text{A.16})$$

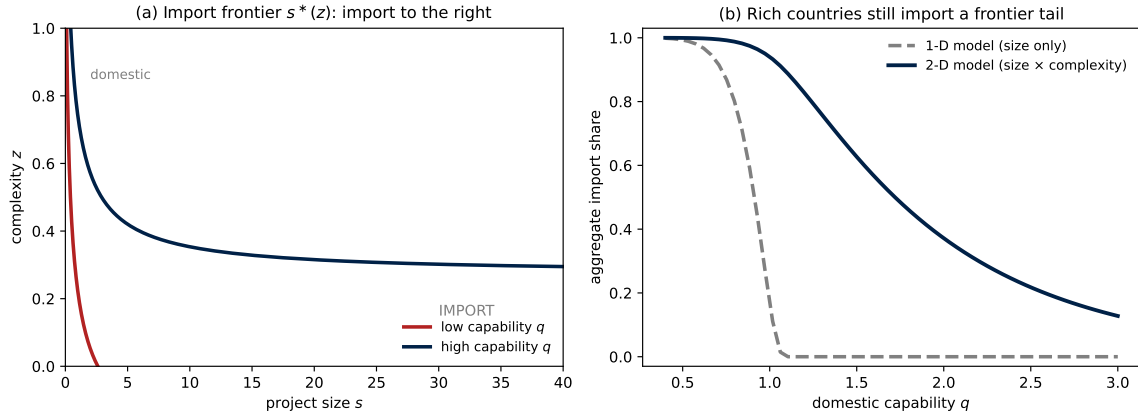
#### M.4. The two-dimensional import set

**Proposition 5** (Sorting in size and complexity). *Define the complexity threshold  $\bar{z}(q)$  by  $a(\bar{z}) = p$ , i.e.  $\bar{z}(q) = g^{-1}(w/(pq))$  (with  $\bar{z} = 0$  if  $w/q \geq p$ ). Then:*

1. *For complexity  $z < \bar{z}(q)$ , domestic engineering is cheaper at every size and all such projects are executed domestically.*
2. *For  $z > \bar{z}(q)$ , projects sort on size exactly as in the baseline: those with  $s > s^*(z)$  are imported, with the size threshold  $s^*(z)$  falling as complexity rises.*
3. *If  $g(z) \rightarrow 0$  as  $z \rightarrow 1$ , then  $a(z) \rightarrow \infty$  and every country with finite capability imports its most complex projects, however large  $q$ .*
4. *A lower  $q$  shifts  $\bar{z}(q)$  down and  $s^*(z)$  in, raising the import share at every complexity; as  $w/q \rightarrow p^+$ ,  $\bar{z} \rightarrow 0$  and the model collapses to the baseline of Section M.1.*

Part 4 is the key consistency check: the extension changes nothing for low-capability countries. When  $w/q \geq p$  the complexity threshold is  $\bar{z} = 0$ , every complexity level is in the size-sorting regime, and the cross-sectional predictions of Section M.2—the binding capacity constraint, the income gradient in the import share—are unchanged. The new content is entirely at the top: capable countries, which the baseline sent to a zero import share, now retain a *frontier tail* of imports concentrated in the most complex projects, regardless of size.

Figure A.12: Frontier expertise and the rich-country import tail



Note: Panel (a): the import frontier  $s^*(z) = \varphi/(a(z) - p)$  in size-complexity space; a project is imported if it lies to the right. Panel (b): the aggregate import share as a function of capability  $q$ . Lognormal sizes;  $g(z) = e^{-\gamma z}$ .

Figure A.12 shows both halves. Panel (a) plots the import frontier  $s^*(z)$  for a low- and a high-capability country: the high- $q$  country imports only in the upper-complexity region, the low- $q$  country across the whole range. Panel (b) contrasts the aggregate import share in the baseline and the extended model as capability rises: the baseline collapses to zero, while the two-dimensional model preserves a positive, complexity-driven import share for even the most capable countries.

## N. The rise of China

We interpret China's entry into global construction as a fall in the world price of engineering services,  $p$ . The supply of internationally tradable engineering expanded, and the price faced by borrowing countries declined.

**Proposition 6** (Effect of a fall in  $p$ ). *A fall in the world price  $p$  raises the aggregate import share through both margins of the model. On the size margin it lowers the sorting threshold  $s^*(z)$  at every complexity (A.16); on the complexity margin it lowers the complexity threshold  $\bar{z}(q)$ , pulling lower-complexity projects into the imported set. Aggregate domestic engineering  $\int_{N_D} s dH$  falls. The fall in  $p$  also lowers the build cutoff  $s_b$ , so more projects are completed and the infrastructure stock  $I$  rises.*

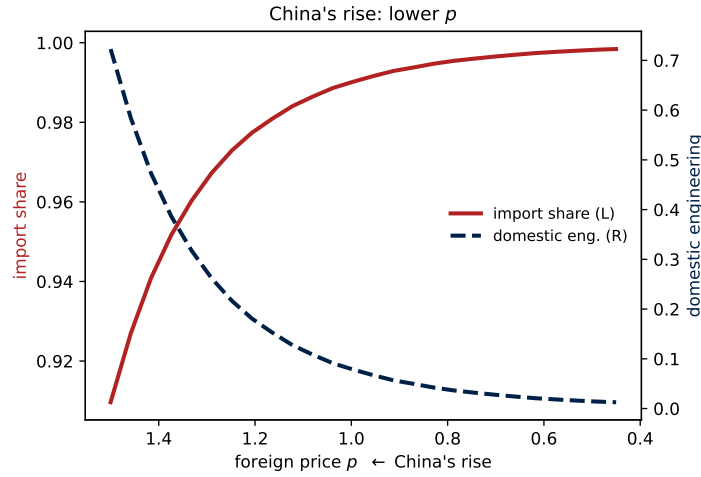


Figure A.13: The rise of China

*Note: A fall in the world price of engineering  $p$  lowers the sorting threshold, raises the aggregate import share (left axis), and crowds out domestic engineering (right axis). Computed from the static equilibrium of Section M.1 with lognormal project sizes.*

The proposition captures Fact 4: cheaper foreign engineering raises the imported share and crowds out domestic engineering, with the projects lost to domestic firms being the largest and most complex ones. The same force reaches capable countries: by lowering  $\bar{z}(q)$ , a fall in  $p$  widens the frontier tail upward into less complex work, so the reallocation toward imports is felt across the whole income distribution, as in the data. At the same time it is unambiguously good for the contemporaneous infrastructure stock—more projects clear the build hurdle. Whether it is good *overall* depends on what happens to domestic capability over time, which is the subject of the next section. Figure A.13 shows the import share rising and domestic engineering falling as  $p$  declines.

## O. Dynamics: learning-by-doing and the cost of importing

The static model takes capability  $q$  as given. We now let it evolve with experience, which turns the static sourcing decision into an intertemporal trade-off and gives imports a dynamic cost.

**Capability accumulation.** Consider two periods  $t = 1, 2$ . Domestic engineering capability in period 2 depends on how much engineering was performed domestically in period 1,

$$q_2 = q_1 + \phi (E_{D,1})^{\chi_L}, \quad E_{D,1} = \int_{\mathcal{N}_{D,1}} s dH(s), \quad \phi > 0, \chi_L \in (0, 1], \quad (\text{A.17})$$

where  $E_{D,1}$  is period-1 domestic engineering volume. Learning is a by-product of executing projects at home: countries that import everything ( $E_{D,1} = 0$ ) accumulate no engineering experience, while countries that retain a domestic engineering base raise their future capability. Concavity ( $\chi_L < 1$ ) is natural and guarantees an interior incentive to keep some engineering at home.

**The planner's trade-off.** The government now maximizes  $u(c_1) + \beta u(c_2)$ . The first period sourcing decision balances a static motive and a dynamic motive. The static motive is the cost comparison of Proposition 2. The dynamic motive is that shifting a project from imports to domestic execution raises  $E_{D,1}$  and hence  $q_2$ , which relaxes the period 2 capacity constraint: a higher  $q_2$  lowers period 2 unit cost  $w_2/q_2$ , raises the period-2 threshold  $s_2^*$ , and lets the country execute more of its future projects at home. Formally the period-1 cutoff becomes

$$s_1^* = \frac{\varphi}{w_1/q_1 - p_1 - \Lambda_1}, \quad \Lambda_1 \equiv \beta \frac{u'(c_2)}{u'(c_1)} \frac{\partial V_2}{\partial q_2} \frac{\partial q_2}{\partial E_{D,1}} \geq 0, \quad (\text{A.18})$$

where  $V_2$  is the period 2 value function and  $\Lambda_1$  is the marginal social value of domestic learning. Relative to the static threshold,  $\Lambda_1 > 0$  raises  $s_1^*$ : the planner keeps more projects at home than static cost-minimization alone would dictate, trading higher current cost for future capability.

**Proposition 7** (Dynamic cost of importing and capability erosion). *If learning is operative ( $\phi > 0$ ) and domestic engineering is interior in period 1, then  $s_1^* > \varphi/(w_1/q_1 - p_1)$ : the country retains more engineering at home than in the static benchmark. A fall in*

*the world price  $p_1$  raises the period 1 import share and lowers  $E_{D,1}$ , which through (A.17) lowers future capability  $q_2$  and, through Proposition 3, raises the future import share.*

Cheaper foreign engineering can therefore reduce a country's long-run engineering capability even as it raises its current infrastructure stock. Proposition 7 is the formal counterpart of the concern that imported expertise “stifles the development of local expertise.” It identifies a genuine trade-off from China's rise. The static gain is cheaper infrastructure today and a larger current stock (Proposition 6). The dynamic cost is reduced domestic learning and a permanently lower capability, which raises future import dependence. The net welfare effect for a given country depends on the strength of learning  $(\phi, \chi_L)$ , the discount factor  $\beta$ , and the project-size distribution  $H$ —the objects a calibration must discipline.

## **P. Policy: taxing foreign engineers**

The dynamic model invites a policy question: should a country tax foreign engineering services to protect its domestic engineering base? We show that the answer, and how it varies across countries, follows from the same sorting logic.

### **P.1. The externality and the role of a tariff**

A tax on foreign engineers has a welfare rationale only when sourcing is decentralized. If the benevolent planner of Section O chooses sourcing directly, it already internalizes learning through the term  $\Lambda_1$  in (A.18), and any tax is a pure distortion. Suppose instead that each project is procured by an atomistic sponsor—a ministry, agency, or developer responsible for a single project—that chooses its engineering source to minimize its own cost, taking current capability  $q_1$  and prices as given. Because each sponsor's project is a negligible part of aggregate domestic engineering, it does not internalize the economy-wide learning spillover in (A.17): learning operates on the aggregate experience stock and is external to the individual project. Domestic engineering is therefore under-provided. The decentralized period-1 cut-

off is the static  $s_{\text{dec}}^* = \varphi/(w/q_1 - p)$ , which omits the learning term and lies below the social cutoff  $\varphi/(w/q_1 - p - \Lambda_1)$ . Appendix ?? sets out the decentralized economy and its implementation in full; the assumption that learning is a pure, uninternalized spillover is what the policy result rests on.

A tax  $\tau$  on foreign engineers raises the price sponsors face to  $p(1 + \tau)$ , with the revenue rebated lump-sum, and moves the private cutoff up:

$$s^*(\tau) = \frac{\varphi}{w/q_1 - p(1 + \tau)}. \quad (\text{A.19})$$

The tax shifts the marginal projects from imports to domestic execution, raising domestic engineering  $E_{D,1}$  and, through (A.17), future capability  $q_2$ .

## P.2. Welfare decomposition and the optimal tariff

Raising  $\tau$  has a static cost and a dynamic benefit. Because the decentralized cutoff is privately cost-minimizing, the static cost is second-order at  $\tau = 0$ : the per-unit resource loss from executing the marginal project at home rather than importing it is exactly  $\tau p$ , which vanishes at  $\tau = 0$ . The dynamic benefit is the uninternalized learning value and the magnitude is first-order. The marginal welfare effect at free trade is therefore unambiguously positive:

$$\left. \frac{dW}{d\tau} \right|_{\tau=0} = \beta \frac{u'(c_2)}{u'(c_1)} \frac{\partial q_2}{\partial E_{D,1}} \underbrace{s_0^* h(s_0^*) \frac{\partial s^*}{\partial \tau}}_{dE_{D,1}/d\tau > 0} > 0. \quad (\text{A.20})$$

**Proposition 8** (Optimal tax on foreign engineers). *With decentralized sourcing and a learning spillover, a small tax on foreign engineers raises welfare. The optimal tax equates the per-unit static cost  $\tau p$  to the marginal social value of domestic learning,*

$$\tau^* = \frac{\beta}{p} \frac{u'(c_2)}{u'(c_1)} \underbrace{\frac{w}{q_2^2} E_{D,2}}_{\text{period-2 cost saving}} \underbrace{\phi \chi_L E_{D,1}^{\chi_L - 1}}_{\partial q_2 / \partial E_{D,1}}, \quad (\text{A.21})$$

where  $w$  is the unit cost of domestic engineering effort and  $w/q$  the cost per efficiency unit. Equation (A.21) is an implicit characterization, since the right-hand side is evaluated at the  $\tau^*$  allocation.

A direct subsidy to domestic engineering attains the same allocation without distorting the import price and is the first-best instrument; the tariff is second-best.

### P.3. How the gains vary across countries

The optimal tax inherits the cross-sectional structure of Section M.2. It is near zero for high-capability countries, which already engineer the bulk of their projects domestically; their residual imports are the frontier tail of Section M.3, which a tariff cannot cheaply repatriate, since domestic cost on those projects ( $a(z)$  large) far exceeds  $p$ . The tax is largest, and the welfare gains greatest, for countries of *low-to-intermediate* capability: those importing enough that protection moves substantial project mass home, yet capable enough that domestic execution is not prohibitively costly. For the very poorest countries the gains turn down again—protection forces engineering onto an extremely high-cost domestic sector ( $w/q_1$  large) and capability accumulates too slowly to repay the static cost within the horizon. The result is a tax schedule  $\tau^*(q_1)$  that declines in initial capability and a hump-shaped profile of welfare gains, illustrated in Figure A.14. Because the frontier tail is imported regardless of the tax, protection builds the routine engineering base rather than frontier capability—reinforcing the distinction drawn in Section M.3.

## Q. Taking the model to the data

The model maps to the three datasets in the empirical section. The project-size distribution  $H$  is identified from the distribution of works contract values in the World Bank procurement data, where the sorting threshold  $s^*$  corresponds to the observed crossover between domestic and imported contracts. The complexity dimension  $z$  and the frontier penalty  $g(z)$  are disciplined by the residual imports of

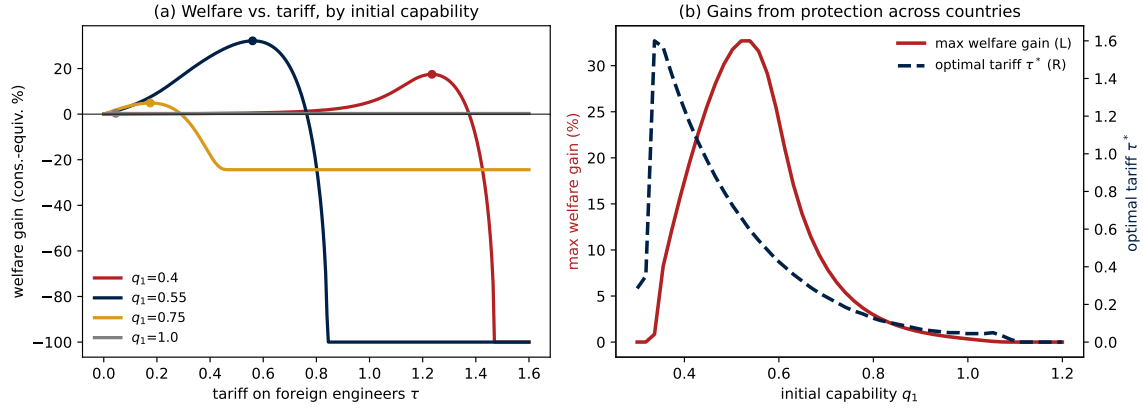


Figure A.14: Welfare effects of a tax on foreign engineers in the decentralized dynamic model. Panel (a): consumption-equivalent welfare gain as a function of the tax  $\tau$ , for countries with different initial capability  $q_1$ ; dots mark the optimum. Panel (b): shows the the optimal tax  $\tau^*$  (right axis) and the welfare gain as a function of initial capability. Lognormal project sizes.

high-capability countries—the sectoral and technical composition of the projects that rich economies still source abroad—using a proxy for  $z$  built from contract sector and structure, analogous to the text-based classification used for the trade data. The import share  $\chi$  and its gradient in income map to the cross-country procurement evidence and pin down the joint distribution of  $(A, q)$  across borrowers together with  $\varphi$ . The decline in  $p$  is disciplined by the rise of Chinese contractors in the ENR data and by the fall in the price of imported construction inputs in the trade data. The learning parameters  $(\phi, \chi_L)$  are identified from the dynamics of domestic contractor participation following episodes of cheaper foreign supply. With these in hand the model quantifies the winners and losers from China’s rise—the static gains to infrastructure-scarce importers against the dynamic losses to countries whose domestic engineering base erodes—and evaluates the policy of Section P, computing the optimal tax on foreign engineers and its welfare consequences across the capability distribution.